

The Gram double pencil: a spectral census for the zeros of ζ , and an Euler-anchored regulator

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the 371.5 rung)

Abstract

We attach to each window of the critical line a pair of Hankel matrices computed by contour integration of ξ'/ξ , and prove that the number of zeros off the critical line in the window equals the negative inertia of the first matrix (Theorem 2); the remaining spectral data of the pair counts the zeros with multiplicity, their distinct support, and their ordinates (Theorem 5). Both statements are unconditional. The Riemann Hypothesis is thereby equivalent to a Stieltjes-type positivity condition, window by window, in a quotient chart obtained from the functional equation (Proposition 6). We then construct, in the half plane of absolute convergence and with no reference to zeros, an Euler-anchored double pencil which is unconditionally positive semidefinite (Proposition 9), and prove that its spectrum can be transported isospectrally by an explicit finite-dimensional regulator whenever a bank response matrix is nonsingular (Theorem 10), extended through spectral collisions by a block regulator (Lemma 12). The census is verified as a computation against the actual zeros, and its detection law is identified: a violation of depth y_0 at height x_0 is seen with strength $2y_0^2 P'(x_0)^2 / \|c\|^2$ (Proposition 45), amplified by the factor $4x_0^2$ in the quotient chart where the functional equation has been used (Proposition 46). We then construct the readout itself: the warp — the flow of a bank-computable velocity field — exists, is real and order-preserving, and carries the frozen labels exactly onto the moving spectrum (Theorem 14); the characteristic polynomial obeys an exact determinant-pullback law with nonvanishing cofactor (Theorem 15). The program thereby reduces to a single hypothesis, the seat (Hypothesis 31): terminal seating of the warped anchor spectrum on the window support, from which the positivity, the census and the critical line follow (Theorem 43). The remaining obligations are stated precisely in §9. A closing section records the operator reading, machine-checked: the functional-equation pairing polarizes the zero multiset as a Krein space whose negative index is the census q and on which the flow generator is unconditionally self-adjoint, and A itself is the generator's spectral determinant — zero multiset equal to the spectrum with multiplicity, canonical genus-one product, two-point resolvent-trace identity (§11).

1 Setup

Let ξ be the completed Riemann zeta function and put

$$A(z) := \xi\left(\frac{1}{2} - iz\right),$$

an entire function of order one, even ($A(-z) = A(z)$, the functional equation), real on $\mathbb{R}$, whose zeros are $z_\rho = \gamma_\rho + i(\frac{1}{2} - \beta_\rho) \cdot (-1)$ for $\rho = \beta_\rho + i\gamma_\rho$ a nontrivial zero of ζ ; in particular

$$\rho \text{ on the critical line} \iff z_\rho \in \mathbb{R}.$$

Nonreal zeros of A occur in conjugate pairs $z, \bar{z}$ with equal multiplicities (conjugate symmetry of ξ on the real axis).

Definition 1 (window moments). Let $W = (a, b)$ with A nonvanishing on $\partial W \times \{|y| \leq Y\}$, $Y > \frac{1}{2}$, and let ∂R_W be the positively oriented rectangle boundary. Put

$$\mu_k(W) := \frac{1}{2\pi i} \oint_{\partial R_W} z^k \frac{A'(z)}{A(z)} dz, \quad k \geq 0,$$

and for $n \geq 1$,

$$H_n(W) := (\mu_{\alpha+\beta}(W))_{\alpha,\beta=0}^{n-1}, \quad H_n^{(1)}(W) := (\mu_{\alpha+\beta+1}(W))_{\alpha,\beta=0}^{n-1}.$$

By the argument principle, with $x_1, \dots, x_r$ the distinct real zeros of A in the window (multiplicities $a_i \geq 1$) and $z_j, \bar{z}_j$, $j = 1, \dots, q$, the distinct conjugate pairs (equal multiplicities $b_j \geq 1$),

$$\mu_k(W) = \sum_{i=1}^r a_i x_i^k + \sum_{j=1}^q b_j (z_j^k + \bar{z}_j^k). \quad (1)$$

All $\mu_k(W)$ are real and computable from A alone; no knowledge of the zeros is used. Write $m := r + 2q$ for the number of distinct support points.

2 The inertia identity

Theorem 2 (inertia). *For every $n \geq m$,*

$$\text{inertia } H_n(W) = (r + q, q, n - r - 2q),$$

where inertia denotes (positive, negative, zero) eigenvalue counts. In particular

$$n_-(H_n(W)) = q, \quad \text{and} \quad H_n(W) \succeq 0 \iff q = 0,$$

i.e. the negative inertia counts exactly the off-critical-line conjugate pairs.

Proof. Let $v(t) := (1, t, \dots, t^{n-1})^T$. By (1),

$$H_n = \sum_{i=1}^r a_i v(x_i) v(x_i)^T + \sum_{j=1}^q b_j \left[v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T \right].$$

Write $v(z_j) = A_j + iB_j$ with $A_j, B_j \in \mathbb{R}^n$; then $v(\bar{z}_j) = A_j - iB_j$ and

$$v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T = 2A_j A_j^T - 2B_j B_j^T, \quad (2)$$

the minus sign being the whole mechanism. Hence $H_n = UJU^T$ with

$$U = [v(x_1) \cdots v(x_r) \ A_1 \ B_1 \cdots A_q \ B_q], \quad J = \text{diag}(a_1, \dots, a_r, 2b_1, -2b_1, \dots, 2b_q, -2b_q),$$

so inertia $J = (r + q, q, 0)$.

U has full column rank m : the complex matrix $V = [v(x_1), \dots, v(x_r), v(z_1), v(\bar{z}_1), \dots]$ is Vandermonde on m distinct nodes with $n \geq m$, so its leading $m \times m$ minor equals $\prod_{\alpha < \beta} (s_\beta -$

$s_\alpha) \neq 0$ and $\text{rank } V = m$; replacing each pair $(v(z_j), v(\bar{z}_j))$ by $(A_j, B_j) = (\frac{v(z_j)+v(\bar{z}_j)}{2}, \frac{v(z_j)-v(\bar{z}_j)}{2i})$ is an invertible change of columns, so $\text{rank } U = m$.

Take a thin QR factorisation $U = QR$ with $Q^T Q = I_m$ and R invertible, and extend Q to an orthogonal $\tilde{Q} = [Q, Q_\perp]$. Then

$$\tilde{Q}^T H_n \tilde{Q} = \begin{pmatrix} R J R^T & 0 \\ 0 & 0_{n-m} \end{pmatrix},$$

and by Sylvester's law of inertia $\text{inertia}(R J R^T) = \text{inertia}(J)$. The zero block contributes $n - m$ null directions. $\square$

Remark 3. The hypothesis $n \geq m$ is necessary: an off-line pair is invisible to a Hankel matrix smaller than the distinct support count. Since $\mu_0(W)$ is the number of zeros in W with multiplicity and $m \leq \mu_0(W)$, the choice $n := \lceil \mu_0(W) \rceil + 1$ is admissible and computable from the bank alone.

Remark 4 (collisions). If a conjugate pair approaches $\mathbb{R}$, the corresponding negative eigenvalue tends to 0^- and at collision becomes a null direction; the invariant through such an event is the inertia, not any individually ordered eigenvalue. Consequently the correct target is non-negativity $H_n \succeq 0$ (equivalently $n_- = 0$), never strict positivity: the $n - m$ structural nulls and the collision nulls are expected.

3 The census, and the equivalent form of RH

Theorem 5 (spectral census). *Let $n \geq m$. From the spectral data of $H_n(W)$ and the value $\mu_0(W)$:*

$$\mu_0 = \text{number of zeros in } W \text{ with multiplicity}, \quad \text{rank } H_n = m = r + 2q,$$

$$n_-(H_n) = q, \quad n_+(H_n) - n_-(H_n) = r,$$

and $\mu_0 > \sum_i 1 + \sum_j 2$ (i.e. μ_0 exceeds the distinct count) if and only if some zero in W is multiple. Moreover the generalized eigenvalues of the pencil $(H_n^{(1)}, H_n)$ restricted to $\text{ran } H_n$ are exactly the support points x_i and $z_j, \bar{z}_j$.

Proof. The first four items are Theorem 2 together with $\mu_0 = \sum a_i + 2 \sum b_j$. For the pencil statement, $H_n = U J U^T$ and $H_n^{(1)} = U J \Sigma U^T$ where Σ is the diagonal matrix of support points in the basis of columns of U (Vandermonde shift identity $v(t)^T \cdot t = \text{shifted moments}$); on $\text{ran } H_n$ the pencil is therefore similar to Σ . This is Gaussian quadrature for the measure (1). $\square$

Proposition 6 (quotient chart; Stieltjes form). *Let $w = z^2$ and $\nu_k(W) := \mu_{2k}(W)$. Under $z \mapsto w$: a real zero maps to $w > 0$; a conjugate pair $x \pm iy$ ($x \neq 0$) maps to a conjugate pair in w ; a pair on the imaginary axis maps to $w < 0$. Consequently, for n large enough in the sense of Theorem 2,*

$$\text{all zeros of } A \text{ in } W \text{ are real} \iff (\nu_{\alpha+\beta}) \succeq 0 \text{ and } (\nu_{\alpha+\beta+1}) \succeq 0,$$

the Stieltjes condition for the measure $\sum_\rho m_\rho \delta_{z_\rho^2}$. Hence RH is equivalent to: for every window of a tiling of $\mathbb{R}$, both Hankel matrices of the even moments are positive semidefinite.

Proof. The functional equation makes A even, so the multiset of zeros is symmetric under $z \mapsto -z$ and descends to the w -chart. Reality and non-negativity of the w -support is exactly the Stieltjes moment condition (Hamburger positivity of both $(\nu_{\alpha+\beta})$ and its shift). The equivalence with the location statement is Theorem 2 applied in the w -chart together with the support map. $\square$

Corollary 7 (sign rigidity). *Let $t \mapsto A_t$ be a continuous deformation with zeros varying continuously and no zero crossing ∂W . Then $H_n(t)$ is continuous, and a conjugate pair is created or annihilated at t_0 if and only if an eigenvalue of $H_n(t)$ crosses 0 at t_0 ; at simultaneous collisions the invariant is the inertia change. Consequently, if $H_n(t) \succeq 0$ for all t , then $n_-(H_n(t)) \equiv 0$ and no off-line pair is ever born. If moreover $H_n(t_*) \succeq 0$ is known at one parameter t_* (for instance in a numerically verified range), the conclusion propagates along the deformation.*

4 Unconditional statements

The following hold with no hypothesis.

- (U1) For every window W and $n \geq m(W)$: $q(W) = n_-(H_n(W))$; the off-critical-line pair count of a window is the negative inertia of a matrix computed by contour integration of A'/A .
- (U2) The census of Theorem 5 holds as stated.
- (U3) RH is equivalent to the Stieltjes condition of Proposition 6 on every window of a tiling, equivalently (Corollary 7) to the absence of a negative-sector eigenvalue crossing under the registered deformation.
- (U4) For every window contained in the numerically verified range ($|t| \leq 3 \cdot 10^{12}$ by the computations of Platt and Trudgian [3]), $q(W) = 0$ and hence $H_n(W) \succeq 0$: the positivity hypothesis is an established fact on an initial segment of the tiling.
- (U5) With $Y > \frac{1}{2}$ the top edge of ∂R_W lies in the half plane of absolute convergence and the bottom edge maps there by the functional equation, giving a decomposition $H_n = H^{\text{pr}} + H^{\text{arch}} + H^{\text{side}}$ into blocks computable from Dirichlet data, Γ -factors and the lateral edges respectively; in a tiling, adjacent lateral contributions cancel pairwise.
- (U6) **The critical abscissa is derived, not posited.** Scale balance asks only that the fiber amplitude reciprocate the arclength radius to a positive finite limit, $n^{-\sigma} \text{carrierRadius}(n) \rightarrow L > 0$; the condition names no zero set. Since $\text{carrierRadius}(n)/\sqrt{n} \rightarrow c > 0$ — the radius being computed from winding at integer sites, with no arithmetic input — the balance product is $(\text{carrierRadius}(n)/\sqrt{n}) n^{1/2-\sigma}$, finite and positive exactly when $\sigma = \frac{1}{2}$. The balanced abscissa is therefore unique and equal to $\frac{1}{2}$ as a theorem of the helix geometry.
- (U7) **The functional-equation pair is area-balanced.** For a nontrivial zero ρ and its reflection $1-\bar{\rho}$, the literal three-dimensional radial magnitudes satisfy $\text{mag}_\rho(n) = n^{-\Re \rho}$ exactly, and

$$\text{mag}_\rho(n) \cdot \text{mag}_{1-\bar{\rho}}(n) = n^{-1},$$

with the area-normalized product tending to 1. No hypothesis on $\Re \rho$ is used. Equality of the two magnitudes holds if and only if $\Re \rho = \frac{1}{2}$.

- (U8) **Boundary viability of the real-rooted locus.** At a double real zero ρ_* the contingent tangent cone to the real-rooted locus is the half space $\{\dot{A} : \dot{A}(\rho_*)A''(\rho_*) \leq 0\}$, and the regulated velocities meet it because the prime channels move A at ρ_* by a nonzero multiple of $p_r^{-\rho_*}$. The locus is viable, and not invariant.
- (U9) **The gate on $[1/4, \infty)$,** with the strict central Turán inequality and the Laguerre–Turán family C_j at every index following from a single gate hypothesis (appendix, §12).

Remark 8 (what is and is not established). The list above is unconditional. The Riemann hypothesis is not among its consequences, and the paper does not claim it: Theorem 21 is stated under Hypothesis 20, Theorem 43 reduces that to Hypothesis 31, and the seat is the single remaining hypothesis (§9). What (U6)–(U8) supply is the geometry around it: the critical line is intrinsically characterised rather than assumed, the functional-equation pair balances collectively, and the boundary of the real-rooted locus is never absorbing.

Combining (U6) and (U7) puts the residue in its sharpest and most nearly one-sided form. Writing $a_n := n^{-\Re \rho} \text{carrierRadius}(n) \sim c n^{1/2 - \Re \rho}$ and b_n for its reflection, one has $a_n b_n \rightarrow 1$ unconditionally, so the pair balances; the hypothesis is that each strand balances individually. Since a_n bounded above forces $\Re \rho \geq \frac{1}{2}$, and since $1 - \rho$ is a zero whenever ρ is, a single upper bound valid at every zero yields $\Re \rho \geq \frac{1}{2}$ and $1 - \Re \rho \geq \frac{1}{2}$, hence $\Re \rho = \frac{1}{2}$. Thus:

$$\text{RH} \iff n^{-\Re \rho} \text{carrierRadius}(n) \text{ is bounded above at every zero.}$$

No lower bound and no two-sided estimate are required — the functional equation supplies the other half — but the upper bound itself is not proved here.

5 The Euler-anchored pencil and its regulator

Nothing in this section refers to zeros; the dependency order is

$$\text{Euler/FE bank} \rightarrow (G_1, G_0) \rightarrow \text{real spectrum} \rightarrow \text{contour transport} \rightarrow \text{residues} \rightarrow \text{zeros},$$

and no step may be inverted.

Proposition 9 (Euler anchor). *Fix $s_0 > 1$ and $N \geq 1$, and set*

$$G_\ell[j, k] := \sum_{n \geq 2} \Lambda(n) n^{-s_0} (\log n)^{j+k+\ell}, \quad 0 \leq j, k < N, \quad \ell \in \{0, 1\}.$$

The series converge absolutely, and G_0, G_1 are the Hankel matrices of the positive measure $\sum_n \Lambda(n) n^{-s_0} \delta_{\log n}$ supported in $[\log 2, \infty)$. Hence unconditionally

$$G_0 \succeq 0, \quad G_1 \succeq 0,$$

and $\text{spec}(G_1, G_0) \subset [\log 2, \infty)$ consists of the Gauss quadrature nodes of that measure.

Proof. Absolute convergence for $s_0 > 1$ is standard. Positivity is the Hamburger/Stieltjes criterion for a positive measure on $[\log 2, \infty)$: for $c \in \mathbb{R}^N$ and $P(x) = \sum_j c_j x^j$, $c^T G_0 c = \int P^2 d\sigma \geq 0$ and $c^T G_1 c = \int x P^2 d\sigma \geq 0$. $\square$

Theorem 10 (unconditional reverb regulator). *Let $t \mapsto (G_0(t), G_1(t))$ be a smooth family with $G_0(t) > 0$ on the active quotient, and set $S(t) := G_0^{-1/2} G_1 G_0^{-1/2}$, so $S = S^*$ and $\text{spec } S = \text{spec}(G_1, G_0)$. Suppose the spectrum of $S(t_0)$ is simple, with spectral projections P_a and eigenvectors v_a , and let the transport be controlled,*

$$\dot{S} = F(u) = F_0 + \sum_{r=1}^m u_r F_r, \quad F_r = F_r^*.$$

Define $M_{ar} := \langle v_a, F_r v_a \rangle$ and $b_a := -\langle v_a, F_0 v_a \rangle$. If $\det M \neq 0$, then $u := M^{-1}b$ is the unique control with $\Delta(F(u)) := \sum_a P_a F(u) P_a = 0$, and with

$$K := \sum_{a \neq b} \frac{P_a F(u) P_b}{\lambda_a - \lambda_b}, \quad K^* = -K,$$

one has $\dot{S} = [S, K]$. Consequently $S(t) = U(t)S(t_0)U(t)^$ with $\dot{U} = -KU$, U unitary, and*

$$\text{spec } S(t) = \text{spec } S(t_0) \quad \text{exactly.}$$

In particular the transported pencil remains Hermitian-definite and its generalized spectrum remains real.

Proof. First-order perturbation theory gives $\dot{\lambda}_a = \langle v_a, F v_a \rangle$, so $\Delta(F)$ is precisely the component of the flow able to move eigenvalues; the linear system $Mu = b$ expresses $\Delta(F(u)) = 0$ and is uniquely solvable iff $\det M \neq 0$. For such u , using $\sum_a P_a = I$,

$$[S, K] = \sum_{a \neq b} (\lambda_a - \lambda_b) \frac{P_a F(u) P_b}{\lambda_a - \lambda_b} = \sum_{a \neq b} P_a F(u) P_b = F(u) - \Delta(F(u)) = F(u) = \dot{S}.$$

$K^* = -K$ since $F(u)^* = F(u)$ and the denominators are real and antisymmetric in (a, b) . The solution of $\dot{U} = -KU$, $U(t_0) = I$, is unitary, and $\frac{d}{dt}(U^* S U) = U^* (\dot{S} - [S, K]) U = 0$. $\square$

Remark 11 (numerical verification). For the Euler anchor of Proposition 9 with $s_0 = 1.5$, $N = 4$, control channels given by reweighting the prime channels $p \in \{2, 3, 5, 7\}$ and raw flow $F_0 = \partial_{s_0} S$: the spectrum is simple $(0.9619, 2.7211, 5.6399, 8.8009)$, $\det M = -0.128$ with $\text{rank } M = 4$ and condition number $3.4 \cdot 10^2$; the solved control annihilates all diagonal drifts to 10^{-12} , and $\|[S, K] - (F - \Delta(F))\| = 1.3 \cdot 10^{-11}$ with $\|K + K^T\| = 1.7 \cdot 10^{-13}$.

Lemma 12 (block regulator through collisions). *Let $S = S^*$ have distinct eigenvalues Λ_α with spectral projections P_α of arbitrary finite multiplicities, and let the flow be controlled as in Theorem 10, $\dot{S} = F(u) = F_0 + \sum_r u_r F_r$, $F_r = F_r^*$. With*

$$\Delta(F) := \sum_\alpha P_\alpha F P_\alpha, \quad K := \sum_{\alpha \neq \beta} \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta},$$

one has $[S, K] = F - \Delta(F)$ and $K^ = -K$. To first order the group Λ_α splits by the spectrum of the compression $P_\alpha F P_\alpha$; hence the regulated condition is the block equation $P_\alpha F(u) P_\alpha = 0$ for all α , a linear system in u generalizing $Mu = b$, and when it is solvable the flow $\dot{S} = [S, K]$ is isospectral with multiplicities. Between collisions Theorem 10 applies; across a collision the invariant is the inertia (Corollary 7); the regulator therefore extends piecewise along any transport with finitely many collision events.*

Proof. $SP_\alpha = \Lambda_\alpha P_\alpha$ and $\sum_\alpha P_\alpha = I$ give

$$[S, K] = \sum_{\alpha \neq \beta} (\Lambda_\alpha - \Lambda_\beta) \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta} = F - \Delta(F),$$

and $K^* = -K$ since $F^* = F$ and the real denominators are antisymmetric in (α, β) . The first-order motion of a degenerate group is degenerate perturbation theory: the group splits by the eigenvalues of $P_\alpha F P_\alpha$, which vanish identically iff the block equation holds; the condition is linear in u , and on solution intervals the Lax argument of Theorem 10 is unchanged. $\square$

6 Warp covariance: the bank-generated readout

Theorem 10 fixes the abstract spectrum $\{\lambda_a\}$ of the regulated flow; the same transport, unregulated, moves it. The first chart makes reality manifest, the second carries the physical ordinates, and the map between them — the warp — is not a hypothesis to be assumed but an object the bank constructs. Throughout this section $t \mapsto (G_0(t), G_1(t))$ is a smooth transport of the continued jet pair, whose inputs are values and derivatives of the continued logarithmic derivative along the path — function data; no zero location enters — and the parameter interval is *regular*: $G_0(t) > 0$ on the active quotient and $\text{spec } S_t$ simple, where $S_t := G_0^{-1/2} G_1 G_0^{-1/2}$. The finitely many failures of regularity are the collision and rank-drop events, crossed piecewise by Lemma 12.

Definition 13 (bank velocity field; warp). Let $\lambda_1(t) < \dots < \lambda_n(t)$ be the eigenvalues of S_t , realized as the generalized eigenpairs $G_1 w_a = \lambda_a G_0 w_a$ normalized by $w_a^T G_0 w_a = 1$, and set

$$d_a(t) := w_a^T (\dot{G}_1 - \lambda_a \dot{G}_0) w_a \quad (= \langle v_a, \dot{S}_t v_a \rangle, \quad v_a = G_0^{1/2} w_a).$$

Let V_t be the unique real polynomial of degree $\leq n-1$ with $V_t(\lambda_a(t)) = d_a(t)$, and let the *warp* φ_t be the flow of the scalar equation $\dot{x} = V_t(x)$, $\varphi_0 = \text{id}$.

Theorem 14 (warp existence and exact covariance). *On every regular interval the warp is defined on a neighborhood of $[\lambda_1(0), \lambda_n(0)]$ and satisfies:*

- (i) **bank determination:** V_t , hence φ_t , is computed from $(G_0, G_1)(t)$ and $(\dot{G}_0, \dot{G}_1)(t)$ alone;
- (ii) **reality and order:** $\varphi_t(\mathbb{R} \cap \text{dom}) \subseteq \mathbb{R}$ and $x \mapsto \varphi_t(x)$ is strictly increasing, in particular injective;
- (iii) **exact covariance:** $\varphi_t(\lambda_a(0)) = \lambda_a(t)$ for every a and every t in the interval.

Proof. (i) is the construction. For (ii): V_t has real coefficients — real nodes, real diagonal drifts of a Hermitian family — so the flow preserves $\mathbb{R}$; the field is polynomial in x with coefficients continuous in t , hence locally Lipschitz in x uniformly on compact t -intervals, so distinct scalar characteristics never cross and φ_t is strictly increasing where defined. For (iii): first-order perturbation theory for the Hermitian-definite pencil gives $\dot{\lambda}_a = w_a^T (\dot{G}_1 - \lambda_a \dot{G}_0) w_a = d_a(t) = V_t(\lambda_a(t))$, so the eigenvalue path $t \mapsto \lambda_a(t)$ solves the initial-value problem defining $\varphi_t(\lambda_a(0))$, and uniqueness makes them equal. Existence: the node characteristics are the globally defined eigenvalue paths; a point between consecutive nodes is trapped between two of them by (ii); and continuous dependence on initial conditions extends the flow to a neighborhood of the compact hull for the same times. $\square$

Theorem 15 (determinant pullback). *Let $p_t(x) := \det(S_t - xI)$, and let I be an open real interval containing the time-0 active spectrum on which the warp is defined. Then on $\varphi_t(I)$,*

$$p_t = J_t \cdot (p_0 \circ \varphi_t^{-1}),$$

with J_t continuous and nonvanishing; consequently

$$Z(p_t) \cap \varphi_t(I) = \varphi_t(\text{spec } S_0) \subset \mathbb{R}.$$

Proof. The field is polynomial in x , so φ_t is a strictly increasing C^1 diffeomorphism of I onto $\varphi_t(I)$. On $\varphi_t(I)$ the functions p_t and $p_0 \circ \varphi_t^{-1}$ have the same zero set $\{\varphi_t(\lambda_a(0))\}_a = \{\lambda_a(t)\}_a$ (covariance and injectivity, Theorem 14), each zero simple (simple spectrum on a regular interval). The ratio $J_t := p_t/(p_0 \circ \varphi_t^{-1})$ is continuous and nonvanishing off the common zeros and extends across each with the nonzero limit $p'_t(\lambda_a(t))/(p_0 \circ \varphi_t^{-1})'(\lambda_a(t))$. $\square$

Remark 16 (the two architectures, reconciled). The regulated and unregulated flows describe one transport in two charts: labels frozen with a moving readout, or a moving spectrum read through the identity; by Theorem 14 the whole discrepancy is φ_t . This corrects the former statement of obligation (N), which asked for a spectral equivalence $\det(G_1 - \lambda G_0) = C \det(H_n^{(1)} - \lambda H_n)$ with one spectral variable λ and a constant C : under the regulated flow that identity is unsatisfiable — the transported spectrum stays at the anchor nodes while the window spectrum sits at the ordinates, disjoint multisets — and it is replaced by the pullback identity of Theorem 15, in which the change of spectral variable is explicit and bank-generated.

Proposition 17 (the warp is internally generated on the anchor segment). *On the segment $s_0 > 1$ the transport is $\dot{\sigma} = -x\sigma$ (each prime clock damps by its own frequency), so $\dot{m}_k = -m_{k+1}$: the moment flow is the pencil's own shift. Let (λ_a, w_a) be the Gauss data of the n -point pencil, P_n the monic node polynomial, and $h_n := \int P_n^2 d\sigma = m_{2n} - \sum_a w_a \lambda_a^{2n}$ the quadrature error of x^{2n} . Then*

$$\frac{d\lambda_a}{ds_0} = -\frac{h_n}{w_a P'_n(\lambda_a)^2},$$

strictly negative for every a . Every quantity on the right except h_n is internal to the pencil; the entire external input of the flow is the single boundary scalar h_n .

Proof. The n -point Gauss data matches the moments m_k for $k \leq 2n - 1$. Differentiating the matching conditions along $\dot{m}_k = -m_{k+1}$, the homogeneous solution $\dot{w}_a = -\lambda_a w_a$, $\dot{\lambda}_a = 0$ accounts for $-m_{k+1}$ within quadrature exactness, leaving the forcing $-h_n \delta_{k,2n-1}$ at the single degree where exactness fails. The correction (u_a, λ_a) solves $\sum_a [u_a Q(\lambda_a) + w_a \lambda_a Q'(\lambda_a)] = -h_n \cdot [x^{2n-1}]Q$ for all Q of degree $\leq 2n - 1$. Take $Q = \ell_a P_n$ with ℓ_a the Lagrange basis: Q vanishes at every node, $Q'(\lambda_b) = \delta_{ab} P'_n(\lambda_a)$, and $[x^{2n-1}]Q = 1/P'_n(\lambda_a)$; the display follows. Positivity of σ on the segment gives $h_n > 0$, $w_a > 0$. $\square$

Remark 18 (self-calibration, and its limit). Proposition 17 is the Toda-deformation structure of exponentially damped measures: the infinite operator is isospectral — its spectrum is the fixed prime-clock support $\{\log n\}$, the carrier — while the finite pencil's nodes, the readout, move only through the boundary forcing h_n . Verified (`tmp/att205.selfcal_probe.py`, $s_0 = 1.5$, $N = 4$): the internal law matches the direct perturbation drifts to $1.6 \cdot 10^{-11}$, with $h_n = 2.2 \cdot 10^5$ cancelling against Christoffel factors of order 10^{-4} exactly. Two consequences and one boundary. First, the

warp needs no external support on the anchor segment beyond one scalar per instant; the continuation's defects (pole, Γ , window) must enter the finite system through this single boundary channel, so Hypothesis 31 becomes a statement about the behavior of a one-dimensional boundary sequence rather than an $n \times n$ identity. Second, $h_n = \int P_n^2 d\sigma$ is manifestly nonnegative exactly as long as the driving functional is a positive measure: where the continuation costs h_n its manifest sign is where the seat's content begins. The boundary: a self-calibrating flow is not a self-certifying one — the device generates its own motion, but the terminal seating certificate must still come from the crossing analysis of Lemmas 23 and 26. Finally, an identity rather than an analogy: for Hankel pairs, $h_n = \det H_{n+1} / \det H_n$, so the boundary scalar that drives the flow *is* the inertia-increment detector of the census — its loss of sign at one size and the appearance of a negative direction at the next are the same fact (verified along the continuation: sign flip at $n = 14$, inertia event at $n = 15$; `tmp/att211_boundary_onset.py`).

Remark 19 (numerical verification). On the convergent segment ($s_0: 1.5 \rightarrow 1.2$, $N = 4$, exact jets of $-\zeta'/\zeta$; `tmp/att201_warp_probe.py`): the perturbation drift matches finite differences of the eigenvalue paths to $8.9 \cdot 10^{-8}$; the flow of the interpolated field carries the anchor nodes (1.1916, 4.0395, 9.6214, 19.3384) onto the terminal eigenvalues (2.1714, 9.2832, 23.2371, 47.5293) with maximal error $1.9 \cdot 10^{-10}$; an interior test point remains strictly interior. The exact-jet anchor nodes differ from those quoted after Theorem 10, which realize the truncated prime sum at finite cutoff; the two banks are distinct legitimate realizations, and the drift between them is the truncation observation of the ledger.

7 The conditional theorem

Hypothesis 20 (PSD). For every window W of a tiling of $\mathbb{R}$ and $n = \lceil \mu_0(W) \rceil + 1$, $H_n(W) \succeq 0$; equivalently (Proposition 6) both Hankel matrices of the even moments are positive semidefinite in the quotient chart.

Theorem 21. *Assume Hypothesis 20. Then*

1. *every nontrivial zero of ζ lies on the critical line;*
2. *for each window the spectral data of $H_n(W)$ returns the full census of Theorem 5, with $n_- \equiv 0$;*
3. *the zeros are the real generalized eigenvalues of the explicitly computed Hermitian pencils $(H_n^{(1)}(W), H_n(W))$.*

Proof. By Theorem 2, $H_n(W) \succeq 0$ with $n \geq m(W)$ forces $q(W) = 0$: no conjugate pair of zeros of A lies in W . Nonreal zeros occur in conjugate pairs with equal real part, hence both members lie in the same window of the tiling (for a pair on a boundary, apply the argument to a shifted tiling). Every nontrivial zero of ζ corresponds to a zero of A lying in some window, and only finitely many per window by Riemann–von Mangoldt. Hence all zeros of A are real, i.e. all nontrivial zeros of ζ have real part $\frac{1}{2}$. Items (2) and (3) are Theorem 5, whose pencil is Hermitian-definite on $\text{ran } H_n$ under Hypothesis 20. $\square$

8 The seat: reduction to a single hypothesis

Definition 22 (terminal defect). Fix a window W with matched dimension n and a transport path whose terminal chart is the window chart, and let $(G_0(1), G_1(1))$ be the terminal

transported pair. Put

$$D_0 := G_0(1) - H_n(W), \quad D_1 := G_1(1) - H_n^{(1)}(W).$$

Both differences are computed from function data — jets of the continued logarithmic derivative on one side, contour integrals of A'/A on the other — so the defect is defined with no reference to zeros. Obligation (T) of §9 evaluates D_ℓ in closed form; the smooth blocks are now explicit. The pole block is the Hankel of the exponential measure $e^{-(s_0-1)x}dx$ on $[0, \infty)$, moments $k!/(s_0 - 1)^{k+1}$, positive semidefinite on the convergent segment $s_0 > 1$. The Γ -block (trivial zeros) is the Hankel of the measure $e^{-(s_0+2)x}(1 - e^{-2x})^{-1}dx$ on $(0, \infty)$, with moments in closed Hurwitz form

$$k! 2^{-(k+1)} \zeta(k+1, 1 + \frac{s_0}{2}) \quad (k \geq 1),$$

a manifestly positive density for every $s_0 > -2$, hence positive semidefinite *throughout the strip*. The entire smooth part of the defect therefore adds positivity at every station of the transport; only the window and lateral blocks of (T) remain unevaluated. (These closed forms are the trivial-zero and pole components of the jet engine used in the numerical maps, validated there against direct differentiation to $5 \cdot 10^{-5}$.)

Lemma 23 (null-cone flow decomposition). *Let $\sigma = \sum_{x \in X} w(x)\delta_x$ be an atomic functional on a conjugation-closed multiset $X = \bar{X}$ with weights satisfying $w(\bar{x}) = \overline{w(x)}$, and let H be the Hankel matrix of its moments, of size n . For every real polynomial P of degree $< n$ with coefficient vector v ,*

$$v^T H v = \sum_{x \in X \cap \mathbb{R}} w(x) P(x)^2 + \sum_{\{z, \bar{z}\} \subset X \setminus \mathbb{R}} 2 \operatorname{Re}[w(z) P(z)^2],$$

in which the real-support sum is termwise nonnegative whenever $w \geq 0$ on $X \cap \mathbb{R}$, while the conjugate-pair terms are unrestricted in sign. Consequently, along a weight homotopy $t \mapsto w_t$ with $\dot{w}_t \geq 0$ on $X \cap \mathbb{R}$, the flow through a null direction ($v^T H_t v = 0$) obeys

$$v^T \dot{H}_t v \geq \sum_{\{z, \bar{z}\}} 2 \operatorname{Re}[\dot{w}_t(z) P(z)^2] :$$

a negative-sector crossing must be fed by conjugate-pair terms; real support cannot produce one. In particular, on any pointwise-monotone kernel homotopy whose weighted support is real and which starts at a nonnegative-weight configuration, $n_-(H_t) \equiv 0$ unconditionally.

Proof. Group the atoms: conjugate pairs combine to $2\operatorname{Re}[w(z)P(z)^2]$ because P is real, and the real-support terms are $w(x)P(x)^2 \geq 0$ termwise. The flow statement is the same identity applied to $\dot{H}_t$, discarding the nonnegative real sum. The last claim is closedness of the PSD cone: positive start, and no crossing is available. $\square$

Remark 24 (measured margin law). On the verified window $W = (10, 32)$ ($n = 5$, four zeros; `tmp/att202_margin_probe.py`), along the error-function smoothing of the window indicator: the dead-direction mass $v_*^T H(\tau) v_*$ (v_* the terminal null polynomial vanishing at the four interior zeros) is monotone with positive flow sign ($+1.4 \cdot 10^{-2}$ at $\tau = 1$); by $\tau = 1$ its source is 100% the boundary-nearest exterior zero $\gamma_5 = 32.935$; and $\lambda_{\min}(H(\tau))$ tracks the Rayleigh value $v_*^T H v_*$ to 99.8% at $\tau = 0.25$. The positivity slack of the window pencil near the sharp limit is the weight leaking onto the nearest exterior zero through the terminal null polynomial: the margin has a single computable source.

Remark 25 (cosine transport of the pair-fed flow). The conjugate-pair terms of Lemma 23 have an exact operator form: for F entire and real on $\mathbb{R}$, $F(x + iy) + F(x - iy) = 2 \cos(y\partial_x)F(x)$ (Taylor expansion, absolutely convergent). Hence a pair at depth y feeds the flow with the *same* real density $g = \dot{w} P^2$ that a real zero would, transported by $\cos(y\partial_x)$. The Fourier multiplier $\cos(y\xi)$ is nonnegative for $|\xi| \leq \pi/(2y)$, and $y < \frac{1}{2}$ unconditionally (the critical strip), so the worst-case positivity radius of the transport is the band $|\xi| \leq \pi$. Two consequences. First, a quantitative target: pair-fed dominance on a null direction requires the density g to carry Fourier mass beyond the positivity radius of the actual pair depths present — shallow pairs feed like real zeros. Second — correcting an identification staked in an earlier draft of this remark — the cross-check against the narrow-support literature *fails on the numbers, and informatively so*: the classical unconditional positivity radius (Yoshida [4]) is $\log 2 = 0.693$, set by the *prime vacuum* (no Λ -support below the first prime), whereas the cosine-transport radius is $\pi = 3.14$, set by pair depth. The two are distinct mechanisms on opposite sides of the explicit formula: prime-vacuum limits the classical regime, pair depth limits this transport, and $\pi > \log 2$ means the pair-depth constraint is strictly the weaker one — positivity of the pair-fed flow persists beyond the prime-vacuum boundary, and the two constraints compose rather than coincide. The present formulation additionally needs the sign only on null directions (a thinner set) and may weigh actual pair depths against unconditional density estimates rather than assuming the worst case everywhere.

Lemma 26 (resonance necessity). *Let P be a real polynomial of degree d with roots $r_1, \dots, r_d$ (nonreal ones in conjugate pairs), let $z_0 = x_0 + iy_0$ with $y_0 > 0$, and let $w(z_0) \in \mathbb{C}^\times$ with $|\arg w(z_0)| \leq \varphi_w < \frac{\pi}{2}$. Define the deviation of each root as the angle between $z_0 - r_j$ and the real line,*

$$\delta_j := \arctan \frac{|y_0 - \operatorname{Im} r_j|}{|x_0 - \operatorname{Re} r_j|} \in (0, \frac{\pi}{2}] \quad (\delta_j := \frac{\pi}{2} \text{ if } x_0 = \operatorname{Re} r_j).$$

If $\operatorname{Re}[w(z_0)P(z_0)^2] \leq 0$, then

$$\sum_{j=1}^d \delta_j \geq \frac{\pi}{4} - \frac{\varphi_w}{2},$$

and in particular some root satisfies

$$|x_0 - \operatorname{Re} r_j| \leq \frac{|y_0 - \operatorname{Im} r_j|}{\tan((\frac{\pi}{4} - \frac{\varphi_w}{2})/d)} \approx \frac{4d}{\pi} (y_0 + |\operatorname{Im} r_j|) \quad \text{for small angles.}$$

Proof. Each squared factor of $P(z_0)^2 = c^2 \prod_j (z_0 - r_j)^2$ contributes, modulo 2π , an argument $\pm 2\delta_j$: for a factor with argument $\theta_j \in (0, \pi)$, either $\theta_j = \delta_j$ or $\theta_j = \pi - \delta_j$, and $2(\pi - \delta_j) \equiv -2\delta_j$. Hence $\arg[P(z_0)^2] \equiv \sum_j \pm 2\delta_j \pmod{2\pi}$, and $|\sum_j \pm 2\delta_j| \leq 2 \sum_j \delta_j$. A nonpositive real part of $w(z_0)P(z_0)^2$ requires its total argument to lie within $\frac{\pi}{2}$ of π modulo 2π , so $2 \sum_j \delta_j + \varphi_w \geq \frac{\pi}{2}$. The root bound is the pigeonhole $\max_j \delta_j \geq (\frac{\pi}{4} - \frac{\varphi_w}{2})/d$. $\square$

Corollary 27 (separation forces depth). *In the setting of Lemma 26, suppose every root satisfies $|x_0 - \operatorname{Re} r_j| \geq d$ and $|\operatorname{Im} r_j| \leq \iota$. Then a nonpositive pair term at (x_0, y_0) requires*

$$y_0 \geq d \tan \frac{\frac{\pi}{4} - \frac{\varphi_w}{2}}{d^\circ P} - \iota \approx \frac{\pi d}{4 d^\circ P} \quad (\varphi_w, \iota \text{ small}),$$

where $d^\circ P$ is the degree. Equivalently: the abscissas from which a pair of depth y_0 can cross are confined to the $((y_0 + \iota)/\tan((\frac{\pi}{4} - \frac{\varphi_w}{2})/d^\circ P))$ -neighborhood of the null-root abscissa set — a set

of total measure at most $\approx \frac{8}{\pi} (d^\circ P)^2 y_0$ — while pairs elsewhere in the window are spectators regardless of the flow.

Proof. Each deviation obeys $\delta_j \leq \arctan((y_0 + \iota)/d)$, so $\sum_j \delta_j \leq d^\circ P \cdot \arctan((y_0 + \iota)/d)$; Lemma 26 requires this to reach $\frac{\pi}{4} - \frac{\varphi_w}{2}$. The neighborhood statement is the same inequality read per root, and the measure bound is $d^\circ P$ intervals of the stated half-width. $\square$

Remark 28 (the composition slot). Corollary 27 is the deterministic half of a two-part exclusion: crossing-capable pairs at depth y are confined to an explicit set of measure $O(n^2 y)$ per window, while the number of pairs of depth $\geq y$ up to height T is counted unconditionally by the zero-density estimates: $N(\sigma, T) \ll T^{3(1-\sigma)/(2-\sigma)+o(1)}$ (Ingham [1]) and $N(\sigma, T) \leq T^{30(1-\sigma)/13+o(1)}$ (Guth–Maynard [2], the first exponent improvement in eighty-four years), so pairs at depth y number $\ll T^{30(\frac{1}{2}-y)/13+o(1)}$. Two calibrations. The composition of the two counts against the window census yields rarity statements — windows admitting a resonance are sparse. Whether that suffices is a quantitative question, not a categorical one: the count of resonant configurations in a given window is a nonnegative integer, so any bound driving it below 1 forces it to 0. What is true is narrower — the exponents currently available do not reach below 1 in the relevant range, since $30(\frac{1}{2} - y)/13 \rightarrow 0$ as $y \rightarrow \frac{1}{2}$ and a bound $T^{o(1)}$ need not be less than one. That is a statement about the present strength of the density estimates, not a barrier to the counting route. Closing the seat on a given window requires the per-window exclusion — path steering within the vulnerable set of Corollary 27, positivity, or a density input sharp enough to cross the integrality threshold — and that is where the remaining content sits.

Remark 29 (crossings are local resonances). Combining Lemmas 23 and 26: a negative-sector crossing requires a conjugate pair at depth $y_0 < \frac{1}{2}$ whose abscissa lies within $\approx \frac{4(n-1)}{\pi} y_0$ of a root cluster of the null polynomial of the accumulated Hankel — with the kernel’s phase budget φ_w a design parameter one makes small on the strip. Shallow pairs can cross only in near-collision with a null root (the continuity of Corollary 7, now quantitative); deep pairs have room, but their number is the subject of unconditional zero-density estimates. The seat’s remaining content is the exclusion of these localized resonances along one designed path per window; the composition of depth, density and the locality of null directions (the measured margin law) is the open program.

Remark 30 (what the local analysis cannot do). Two of the natural attacks on that remaining content are foreclosed, and it is worth recording why so the effort is not spent. At a *null direction* with real support the relevant eigenvalue reaches zero with second derivative $((2k)!/(k!)^2) \sum_\rho w_\rho (P'(\rho) \rho^{(k)})^2 \geq 0$, a sum of squares: the determinant touches and does not cross. That mechanism does *not* extend to a node *collision*. In the minimal model — two unit nodes at $\pm s$, so $\det H = m_0 m_2 - m_1^2 = 4s^2$ — the natural parameter through a merge is $\tau = s^2$, in which $\det H = 4\tau$ is *linear*: the crossing is transversal, so no statement about the local order of vanishing can exclude the off-axis branch. (Consistently, a conjugate pair $w\delta_z + \bar{w}\delta_{\bar{z}}$ gives $\det H = -4y^2(u^2 + v^2) < 0$ exactly.)

Nor can the collision be avoided by choosing the path better. For a real monic polynomial $\text{sign}(\text{disc}) = (-1)^{r_2}$ with r_2 the number of conjugate pairs, and disc is a polynomial in the coefficients, hence continuous along any path; passing from $r_2 = 0$ to $r_2 = 1$ therefore forces $\text{disc} = 0$ somewhere, which is precisely a repeated node. The collision wall is thus codimension one *within the moment family* — the single condition $\rho_i = \rho_j$ — rather than the codimension two of eigenvalue coincidence in an unconstrained real-symmetric family, so genericity buys nothing: any transport whose target carries a conjugate pair must cross it. What happens at

that crossing is therefore decided by which side of the wall the zeta configuration lies on, which is the conclusion. This delimits the construction, not the problem: it says the seat cannot be closed by local analysis at its degeneracies, and that an argument for it must be global or arithmetic.

Hypothesis 31 (the seat (S)). For every window W of the tiling there is a transport path with finitely many regularity events such that every point of $\text{spec}(H_n^{(1)}(W), H_n(W))|_{\text{ran } H_n}$ — the window support — is a terminal limit of the warped anchor spectrum $\varphi_t(\text{spec } S_0)$, the remaining $n - m$ directions retiring into the null space through the rank-drop events.

Lemma 32 (bounded bank field; global warp). *The polynomial choice in Definition 13 is inessential, and replacing it globalizes the warp. Let $\psi_a \in C^\infty(\mathbb{R})$ satisfy $\psi_a \geq 0$, $\sum_a \psi_a \equiv 1$ and $\psi_a(\lambda_b) = \delta_{ab}$ — available as a partition of unity subordinate to the node gaps, the nodes being separated on a regular interval, and fixed canonically by scaling to the gaps. Put $V_t := \sum_a d_a(t) \psi_a$. Then $V_t(\lambda_a) = d_a$, V_t is real and C^∞ , and*

$$\|V_t\|_\infty \leq \max_a |d_a(t)|,$$

since V_t is a convex combination of the d_a . Consequently, on a regular interval, the flow of $\dot{x} = V_t(x)$ is complete,

$$|\varphi_t(x) - x| \leq \int_0^t \|V_s\|_\infty ds \leq \int_0^t \max_a |d_a(s)| ds,$$

and φ_t is a C^∞ diffeomorphism of $\mathbb{R}$ onto $\mathbb{R}$.

Proof. Interpolation and the sup bound are immediate from $\psi_a \geq 0$, $\sum \psi_a = 1$. A bounded, locally Lipschitz field has no finite-time blow-up, so the flow is complete; strict monotonicity is as in Theorem 14(ii), and $\varphi_t(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$ by the displacement bound, so φ_t is onto. $\square$

Two caveats, both load-bearing. The partition subordinate to the node gaps degenerates as $\lambda_a - \lambda_b \rightarrow 0$: boundedness of V_t survives a collision, since it is a convex combination throughout, but the Lipschitz constant does not, and with it uniqueness and monotonicity. The lemma is therefore stated on a regular interval, exactly as Theorem 14 is, and it does *not* by itself carry the warp across a collision. Second, the globalization must not be over-read: S_t is Hermitian along the regulated flow, so $\text{spec } S_t \subset \mathbb{R}$ automatically and p_t is real-rooted for trivial reasons. Theorem 15 was never establishing reality; it establishes the *identification* $Z(p_t) = \varphi_t(\text{spec } S_0)$, which is covariance. What the bounded field buys is that this identification holds on all of $\mathbb{R}$ rather than on a neighborhood of the initial spectrum — a removal of a domain restriction, not a gain in arithmetic content.

All three clauses of Theorem 14 are untouched: (i) the field is still computed from $(G_0, G_1, \dot{G}_0, \dot{G}_1)$; (ii) reality and order need only real coefficients and local Lipschitz continuity; (iii) is the interpolation condition. In Theorem 15 the interval I may therefore be taken to be all of $\mathbb{R}$, strengthening its conclusion to $Z(p_t) = \varphi_t(\text{spec } S_0) \subset \mathbb{R}$ globally, and the “neighborhood of $[\lambda_1(0), \lambda_n(0)]$ ” of Theorem 14 becomes $\mathbb{R}$. The polynomial performed no function except to bound the domain; its degree is also what carries the factor $n - 1$ in Remark 29 and n^2 in Corollary 27, so a bounded field shrinks the vulnerable set as well.

Lemma 33 (null directions retire; sum of squares). *Let $G_0(t) = \sum_{\rho} w_{\rho}(t) v(\rho(t)) v(\rho(t))^T$ with real nodes and $w_{\rho} > 0$, and let w span a null direction, P the associated polynomial. Then $w^T G_0 w = \sum_{\rho} w_{\rho} P(\rho)^2 = 0$ forces $P(\rho) = 0$ at every node, whence $w^T \dot{G}_0 w = 0$ and, with k the least order for which $\rho^{(k)} \neq 0$,*

$$\left. \frac{d^{2k}}{dt^{2k}} \right|_0 w^T G_0 w = \frac{(2k)!}{(k!)^2} \sum_{\rho} w_{\rho} (P'(\rho) \rho^{(k)})^2 \geq 0.$$

A null direction therefore retires into the null space and never crosses into the negative sector; equivalently, and more directly, $w^T G_0(t) w = \sum_{\rho} w_{\rho}(t) P(\rho(t))^2 \geq 0$ identically. The frozen case $\dot{\rho} = 0$ needs no separate treatment.

The hypothesis of Lemma 33 is the real-support, positive-weight regime, which is precisely the regime whose persistence is at issue; the lemma is therefore a statement about what can happen *while* that regime holds, and cannot on its own establish that it persists through the first loss of reality. Read correctly, and combined with Corollary 7 — a conjugate pair is created exactly when an eigenvalue of H_n crosses 0 — it says that so long as the support is real the crossing cannot originate at a null direction. Pair creation must therefore occur at a collision of two real nodes splitting off-axis, which near a double zero ρ_* of A_{t_0} is the local sign condition

$$(z - \rho_*)^2 = - \frac{2\dot{A}(\rho_*) (t - t_0)}{A''(\rho_*)}, \quad \text{real split} \iff \dot{A}(\rho_*) A''(\rho_*) \leq 0.$$

Proposition 34 (viability of the real-rooted locus). *Let R denote the set of configurations all of whose window zeros are real. At a boundary point $A_0 \in \partial R$, carrying a double real zero at ρ_* , the contingent tangent cone is the closed half-space*

$$T_R(A_0) = \{ \dot{A} : \dot{A}(\rho_*) A''_0(\rho_*) \leq 0 \},$$

since $A_t(z) = \frac{1}{2} A''_0(\rho_) (z - \rho_*)^2 + t \dot{A}(\rho_*) + o(\cdot)$ has real zeros near ρ_* for $t > 0$ exactly under that inequality. The velocities realizable by the regulated flow $\dot{S} = F_0 + \sum_r u_r F_r$ form an affine set $\mathcal{A}$, and the cone is cut by the single linear functional $\ell(\dot{A}) = \dot{A}(\rho_*) A''_0(\rho_*)$. Hence $\mathcal{A} \cap T_R(A_0) \neq \emptyset$ — Nagumo's condition — unless ℓ is strictly positive on all of $\mathcal{A}$, which fails as soon as one control direction has $\dot{A}_r(\rho_*) \neq 0$.*

For the prime channels $p \in \{2, 3, 5, 7\}$ of Remark 11, the r -th direction moves A at ρ_ by a multiple of $p_r^{-\rho_*}$, which is nonzero for every finite ρ_* . The non-degeneracy therefore holds at every collision site with no hypothesis, and R is viable: from any real-rooted configuration, and at every time along the transport, there is an admissible velocity pointing into R .*

Corollary 35 (viable but not invariant). *R is not invariant under the regulated flow. Invariance would require $\mathcal{A} \subseteq T_R(A_0)$, i.e. $\ell \leq 0$ on every realizable velocity; but the same non-vanishing $p_r^{-\rho_*} \neq 0$ that supplies viability lets the controls realize $\dot{A}(\rho_*)$ of either sign, so $\mathcal{A}$ straddles the hyperplane $\ell = 0$.*

This dichotomy is exactly why Hypothesis 31 is existential. Were R invariant, reachability would be automatic: the anchor lies in R , every trajectory would remain there, and the terminal object would inherit $q = 0$ with no choice to make. Because R is merely viable, staying inside is always possible but never forced, and the seat is precisely the assertion that the possibility can be exercised all the way to the window. The content of the seat is thus located exactly: not in the

boundary geometry, which is now settled, but in whether the admissible choices can be made consistently along a route terminating at the given window.

The boundary of the real-rooted locus is thus never absorbing — the failure mode that Lemma 23 and the collision analysis were circling is excluded outright. What viability does *not* supply is reachability: it produces trajectories that remain in R , not one that attains a prescribed target. The window's Hankel is fixed data and the control selects the route rather than the endpoint, so a route from the anchor to the window inside R exists precisely when the window lies in R , i.e. when $q(W) = 0$. Viability removes the obstruction along the way; the endpoint remains.

Remark 36 (what the collision criterion leaves). The three statements above remove, respectively, the domain obstruction on a regular interval, the null-direction obstruction while the real-support regime holds, and the generic-collision obstruction. What remains is the sign condition $\dot{A}(\rho_*)A''(\rho_*) \leq 0$ at each collision event. Here $A''(\rho_*)$ is fixed by the local shape and $\dot{A}(\rho_*)$ is a design parameter of the deformation, so the seat is a *construction*: exhibit a transport whose velocity carries the correct sign at every collision.

It should be recorded that path-independence of $n_-(H_n(W)) = q(W)$ does *not* argue against this. That the answer is independent of the route is what makes a route-based proof transfer, not what forbids one; a crossing-free path to the actual window exists precisely when $q(W) = 0$, so exhibiting one is a proof rather than a verification. The cost is that the construction cannot be checked more cheaply than the theorem it establishes — but unlike an estimate, it fails concretely, at a nameable collision, rather than by an insufficient margin. For a problem with no margin to spare that is the appropriate species of argument.

Remark 37 (three contractions of one kernel). The transported objects of this paper are contractions of the single two-variable kernel $B_{jk}(\rho, \rho') = (s_0 - \rho)^{-(j+1)}(s_0 - \rho')^{-(k+1)}$ along three pairings of the conjugation- and FE-closed zero multiset: the *holomorphic diagonal* (ρ with ρ ; the jet-Hankel face, real symmetric), the *functional-equation pairing* (ρ with $1 - \rho$; hermitian by conjugation-closure), and the *conjugate pairing* (ρ with $\bar{\rho}$; a Gram matrix, positive semidefinite always). Atomwise, the FE and conjugate pairings coincide iff $\bar{\rho} = 1 - \rho$, i.e. iff the zero lies on the critical line; the holomorphic diagonal is indefinite for complex frequencies whether or not they lie on the line. Two consequences. First, a calibration: the indefiniteness onset of the jet-Hankel face along the continuation carries no information about zero locations — it is the generic behavior of complex-frequency moment data. Second, the diagnostic content of the program is exactly the coincidence of the FE and conjugate contractions — every zero self-paired by the functional equation is a zero on the line — which is the pairing-language form of the single hypothesis, and the positivity of the FE contraction is its window-free shadow.

Proposition 38 (scalar seat criterion). *Over the nontrivial-zero multiset put*

$$S(s) := \sum_{\rho} (s - \rho)^{-1} (\bar{s} - 1 + \rho)^{-1},$$

absolutely convergent for s off the zero set (terms are $O(|\gamma_{\rho}|^{-2})$), and real-valued: the relabeling $\rho \mapsto 1 - \bar{\rho}$, under which the multiset is closed, conjugates its terms. Then

every nontrivial zero lies on the critical line $\iff S(s) \geq 0$ for all s in the strip off the zero set.

Proof. ($\Rightarrow$) For ρ on the line, $1 - \rho = \bar{\rho}$, so the ρ -term is $(s - \rho)^{-1} \overline{(s - \rho)^{-1}} = |s - \rho|^{-2} > 0$: the sum is termwise positive. ($\Leftarrow$) Let $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac{1}{2}$ and set $s = \rho_0 + \varepsilon e^{i\theta}$. The

ρ_0 -term is $\varepsilon^{-1}e^{-i\theta}((2\beta_0 - 1) + \varepsilon e^{-i\theta})^{-1}$, of modulus $\sim (\varepsilon|2\beta_0 - 1|)^{-1}$ with freely rotating phase; choosing θ with $\cos \theta \cdot (2\beta_0 - 1) < 0$ makes its real part $\rightarrow -\infty$ as $\varepsilon \rightarrow 0$. Every other term stays bounded on the ε -ball: the remaining members of the ρ_0 -family ($\bar{\rho}_0, 1 - \rho_0, 1 - \bar{\rho}_0$) and all other zeros are at distance bounded below, and the full sum of their moduli converges. Hence $S(s) < 0$ for small ε . $\square$

Remark 39 (the traveling seat). Proposition 38 is the $n = 1$ member of the FE-paired family $A_{jk}(s) = \sum_{\rho} (s - \rho)^{-(j+1)}(\bar{s} - 1 + \rho)^{-(k+1)}$, Hermitian for every anchor s by the same relating, with on-line zeros contributing rank-one positive terms at every anchor. Numerically (`tmp/att217_anchor_lift.py`, verified range, $n = 8$): as the anchor rises through the strip the form remains positive semidefinite at every station and *seats* on each zero in turn — mass share of the nearest zero 99% at $\tau = \gamma_j$, resolvent spike at each seating, conditioning improving with localization. A negative direction would appear exactly at the seating of an off-line pair: the falsifier is localized in the anchor height.

The source reading required by the working rules identifies the scalar criterion: by partial fractions, $(s - \rho)^{-1}(\bar{s} - 1 + \rho)^{-1} = (s + \bar{s} - 1)^{-1}[(s - \rho)^{-1} + (\rho - (1 - \bar{s}))^{-1}]$, and summing over the paired multiset via the Hadamard expansion and the functional equation ($\xi'/\xi(1 - \bar{s}) = -\bar{\xi}'/\bar{\xi}(\bar{s})$) gives the closed form

$$S(s) = \frac{2 \operatorname{Re} [\xi'/\xi(s)]}{2\sigma - 1}, \quad \sigma = \operatorname{Re} s.$$

Proposition 38 is therefore equivalent to the positivity criterion of Hinkkanen (1997) and Lagarias [5]: RH iff $\operatorname{Re} [\xi'/\xi](s) > 0$ for $\sigma > \frac{1}{2}$ — a rediscovery, in FE-paired coordinates, with the pairing trichotomy of Remark 37 supplying a two-line derivation. What is retained beyond the identification: the anchor-lift geometry above (the criterion's falsifier is seat-localized), and the wiring of the criterion into the transport apparatus of this paper. The graded family A_{jk} for $n > 1$ is likewise identified by the required source reading: it is of the Weil–Yoshida hermitian-form species [4, 7] in a resolvent basis at complex anchor — the species carrying Li's criterion as its Laguerre-basis instance (Bombieri–Lagarias [6]) and the Connes–Consani semi-local positivity results as its deepest partial positivity. The bridge runs both ways: partial positivity results for these forms are partial results on the seat, and the transported-pencil apparatus — the internal flow law, the boundary-scalar detector, the seat-localized falsifier geometry — is what this paper adds around the classical species.

Proposition 40 (unconditional positivity on the verified band). *Let T_0 be a height below which every nontrivial zero lies on the critical line ($T_0 = 3 \cdot 10^{12}$ by [3]). There is an absolute constant C such that*

$$S(s) > 0 \quad \text{for every } s = \sigma + i\tau \text{ in the strip with } 30 \leq |\tau| \leq T_0 - C \log T_0.$$

The scalar seat criterion of Proposition 38 therefore holds unconditionally on the full anchor band of the verified range; its open content is confined to anchors above the verification height.

Proof. Split the sum at height T_0 . Below: every zero is on the line, so each term is $|s - \rho|^{-2} > 0$; by Riemann–von Mangoldt, $[|\tau|, |\tau| + 1]$ contains a zero once $|\tau| \geq 30$, giving a single-term lower bound $S_{<}(s) \geq (\frac{1}{4} + 1)^{-1} \cdot \frac{1}{2}$, say. Above: for any zero ρ (on the line or not) at height $\gamma > T_0$, both factors have modulus at least $\gamma - |\tau|$ — $|s - \rho| \geq \gamma - |\tau|$ and $|\bar{s} - 1 + \rho| \geq \gamma - |\tau|$, since each has imaginary part $\pm(\gamma - |\tau|)$ up to sign conventions — so the term is bounded by $2(\gamma - |\tau|)^{-2}$

in modulus. Integrating against the counting measure,

$$|S_{>}(s)| \leq 2 \int_{T_0}^{\infty} \frac{dN(\gamma)}{(\gamma - |\tau|)^2} \ll \frac{\log T_0}{T_0 - |\tau|},$$

which is smaller than the lower bound once $T_0 - |\tau| \geq C \log T_0$ for an absolute C . $\square$

Proposition 41 (band-local seat criterion). *For $H \geq 60$, with C the absolute constant of Proposition 40:*

- (i) *if every zero with $|\gamma| \leq H$ lies on the line, then $S(s) > 0$ for all anchors with $30 \leq |\tau| \leq H - C \log H$ (the proof of Proposition 40 verbatim, with T_0 replaced by H);*
- (ii) *if $S \geq 0$ on the band $|\tau| \leq H$, then every zero with $|\gamma| \leq H - 1$ lies on the line (the converse construction of Proposition 38 produces a negative value at an anchor within distance 1 of any off-line zero).*

In particular RH is equivalent to band positivity at every height, with the correspondence exact up to logarithmic boundary layers.

Remark 42 (the induction frame). Proposition 41 places the whole question in one-dimensional coordinates: RH holds iff the band-positivity property is unbounded, and by (U4) it holds below $3 \cdot 10^{12}$. The single remaining statement is the self-extension step — band positivity below H implies it below $H + C \log H$ — and by (ii) this step is equivalent to the absence of a first off-line zero in the boundary layer: necessarily of full strength, as every reduction in this paper must be, but now localized to a moving layer of logarithmic width, where the protecting mass is the verified band below and the threat is a single hypothetical seating just above. The seat, in its minimal coordinates: *the boundary layer never turns.*

Theorem 43 (the chain). *Hypothesis 31 implies Hypothesis 20. Hence, granted the seat alone: every nontrivial zero of ζ lies on the critical line, the census of Theorem 5 is returned on every window, and the zeros are the real spectra of the computed Hermitian pencils (Theorem 21).*

Proof. On every regular interval the transported spectrum is real: the anchor spectrum is real (Proposition 9, S_0 Hermitian) and Theorem 14 transports it by a real map; across regularity events reality is preserved because the pencil remains Hermitian with $G_0 > 0$ on the active quotient (Lemma 12). A limit of reals is real, so by Hypothesis 31 every support point of the window measure (1) is real: $q(W) = 0$ for every window, and Theorem 2 gives $H_n(W) \succeq 0$: Hypothesis 20. $\square$

Remark 44 (necessity, and what has been gained). By the rigidity of the moment problem (§9) any hypothesis implying Hypothesis 20 is equivalent to it; the seat is no exception, and this is forced — it is not a defect of the method. What has changed is the form: of the former obligations, (N), (W1), (W2) and (C) are now theorems (Theorem 10, Lemma 12, Theorems 14 and 15), and the residue is one statement, bank-generated and zero-free in formulation, finite-dimensional on each window.

9 Remaining obligations

Theorem 21 is conditional on Hypothesis 20, and Theorem 43 reduces Hypothesis 20 to the seat. By the rigidity of the moment problem, any pairing whose Gram matrix reproduces the moments $\mu_{j+k}(W)$ is the counting pairing of (1), so its positivity is equivalent to the conclusion. Equivalence is not circularity: either side may be attacked, and a direct proof of positivity would be a proof of the conclusion rather than an appeal to it. What the rigidity does say is that no *cheap* proof of positivity can exist — one cannot verify it by exhibiting some other pairing with the same moments, since there is no other. The bank side is therefore one route among the available ones, taken here because it supplies structure the moment side does not; the precise remaining steps are these.

- (T) **Contour transport (defect evaluation).** Evaluate the terminal defect of Definition 22 in closed form,

$$G_\ell[j, k] = \frac{1}{2\pi i} \oint_\Gamma z^{j+k+\ell} \frac{A'(z)}{A(z)} dz + D_\ell[j, k],$$

with D_ℓ the explicit pole, Γ -factor, lateral and windowing contributions of (U5). The chain of Theorem 43 does not consume (T): the defect is already defined as a difference of computed objects; (T) supplies its closed form, the DC structure of the seat, and computational access to Hypothesis 31 on explicit windows.

- (S) **The seat.** Hypothesis 31: terminal seating of the warped anchor spectrum on the window support. This is the single remaining hypothesis; its channel-localized equivalent form, with a machine-checked certified floor to height 743/2, is §11.1. By Lemma 23 its content along monotone kernel homotopies is exactly: the conjugate-pair-fed component of the flow never dominates the real-fed component on a null direction of the accumulated Hankel — the only crossing mechanism that exists. The former obligations (N), (W) and (C) are discharged: pencil neutrality by Theorem 10 with the corrected terminal statement of Remark 16; warp covariance — bank determination, reality, injectivity, and the determinant-pullback law — by Theorems 14 and 15; collisions by Lemma 12.

Granted (S), the chain runs forward: the Euler-anchored pencil is positive semidefinite unconditionally (Proposition 9); its spectrum is real and is carried by the bank-generated warp along the transport, exactly and order-preservingly (Theorems 10, 14, 15, Lemma 12); the seat identifies the terminal limits with the window support; reality forces $q \equiv 0$ (Theorem 2); and Theorem 21 concludes.

10 The census as a computation: verification, resolution law, and the functional-equation gain

The census of §9 is not only a theorem; it is an algorithm, and its behaviour has been measured. Everything in this section was computed for the window $W = (10, 30)$ with rectangle height $Y = 3/2$, against the three zeros 14.134725, 21.022040, 25.010858 it contains (`tmp/att220-att233`).

10.1 The contour blocks, evaluated

Decompose ∂R_W into the two horizontal (windowing) edges and the two lateral edges.

- (a) **Laterals telescope exactly.** The lateral moments at $x = 10, 20, 30$ were computed directly from A'/A ; the shared edge of two adjacent windows carries identical values with opposite orientation, so adjacent contributions cancel to zero at every k . This verifies the pairwise cancellation asserted in (U5).
- (b) **The windowing block is prime-computable — through the functional equation.** The top edge lies at $\operatorname{Re} s = \frac{1}{2} + Y = 2$, in the half plane of absolute convergence, and recomputing its moments from

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{s}{2}\right) + \frac{\zeta'}{\zeta}(s), \quad \frac{\zeta'}{\zeta}(s) = - \sum_{n \geq 2} \Lambda(n) n^{-s},$$

with no zero data whatsoever, reproduces the contour values to relative $2.3 \cdot 10^{-6}$ – $2.7 \cdot 10^{-7}$. The bottom edge, by contrast, lies at $\operatorname{Re} s = \frac{1}{2} - Y = -1$, where that series *diverges*: it is accessible only as the functional-equation image of the top edge, and the two were measured to be exactly equal. The functional equation is therefore load-bearing in the prime-side representation, not decorative.

- (c) **Signs.** The pole and Γ blocks are positive semidefinite (proved; §8), the laterals telescope, and the windowing block alone is *indefinite* — inertia $(2, 1)$ at $n = 3$. The assembled matrix is nevertheless positive semidefinite: a summand fails positivity while the sum does not.

10.2 Verification against the actual zeros

Assembling all blocks reproduces the atomic prediction $\mu_k = \sum_i x_i^k$ to 8–10 digits ($\mu_0 = 3$ exactly by the argument principle; $\mu_1 = 60.16762236$ against 60.167623 ; $\mu_5 = 1.445663 \cdot 10^7$ against $1.4456633 \cdot 10^7$). The census returns inertia $(n, 0)$, $q = 0$, with the structural rank drop appearing exactly at $n = 4$ as Theorem 2 requires for three distinct support points.

10.3 The functional equation and the shape of a violation

A is even, $A(-z) = A(z)$ — this is the functional equation in the z -chart. Consequently a genuine off-critical zero appears not as a conjugate pair but as a *quadruple* $\{\pm x \pm iy\}$, and a synthetic test that plants only $\{w, \bar{w}\}$ is not a ξ -like object. The functional-equation-respecting formulation is the quotient chart $w = z^2$ of Proposition 6, in which the quadruple folds to a single conjugate pair and the test is the Stieltjes pair condition. Measured there at the size threshold $m = 5$: the true configuration gives $n_- = 0$ in both Hankels $(-1.9 \cdot 10^{-53}, -7.7 \cdot 10^{-57})$, an off-line quadruple at $18 \pm 0.01i$ gives $n_- = 1$ in both $(-3.2 \cdot 10^{-8}, -1.0 \cdot 10^{-5})$.

Proposition 45 (resolution law). *Let P be a real polynomial and let a conjugate pair sit at $x_0 \pm iy_0$. Its contribution to the quadratic form $c^T H_n c = \sum_\rho P(z_\rho)^2$ is*

$$2 \operatorname{Re}[P(x_0 + iy_0)^2] = 2P(x_0)^2 - 2y_0^2[P'(x_0)^2 + P(x_0)P''(x_0)] + O(y_0^4).$$

Hence the detecting direction is the P vanishing at the real support and at x_0 , and the detected eigenvalue is

$$\lambda_{\min} = -\frac{2y_0^2 P'(x_0)^2}{\|c\|^2} + O(y_0^4) :$$

the depth enters as y_0^2 and the position through a Christoffel-type factor.

Measured against Proposition 45 with no fitted parameter, at $y_0 = 10^{-2}$, $n = 5$: ratios of measured to predicted are 1.0001, 1.0002, 1.0001, 1.0000, 1.0001, 1.0002, 1.0002 at $x_0 = 11, 12, 16, 18, 23, 28, 29$ — the law reproduces a three-order spread in the signal exactly; and 1.00017, 1.00004, 1.00001 across $y_0 = 2 \cdot 10^{-2}, 10^{-2}, 5 \cdot 10^{-3}$. When x_0 coincides with a support point the detecting polynomial satisfies one fewer vanishing condition, the extremal direction changes, and the signal is far larger — a second, equally explicit regime.

Two consequences. First, a working-precision requirement: detecting a pair of depth y_0 costs

$$\text{digits} \approx \underbrace{\log_{10} \kappa}_{\text{dynamic range}} + 2 \log_{10}(1/y_0) + \log_{10}(\|c\|^2/P'(x_0)^2),$$

so the census is an unconditional theorem with an algorithm attached rather than an effective verification method: at double precision the detection fails at every depth tested, including $y_0 = 0.3$.

Second, and in the other direction:

Proposition 46 (functional-equation gain). *In the quotient chart $w = z^2$ a violation at $x_0 \pm iy_0$ sits at $w = (x_0 + iy_0)^2$ with*

$$\text{Im } w = 2x_0y_0,$$

so folding by the functional equation multiplies the effective depth by the height. By Proposition 45 the detected signal gains a factor $4x_0^2$.

Measured at the same physical violation: $\times 440$ at $x_0 = 18$ ($-7.2 \cdot 10^{-11}$ in the z -chart against $-3.2 \cdot 10^{-8}$ in the w -chart) and $\times 2300$ at $x_0 = 30$. Detection sensitivity therefore improves quadratically with height in the functional-equation chart, against the degradation of the raw chart — the symmetry buys the resolution.

10.4 Where the positivity comes from

On the near-null direction of the assembled matrix at $n = 3$, splitting the energy into its prime, archimedean and boundary parts gives $-1.9 \cdot 10^{-3}$, $+3.0 \cdot 10^{-2}$ and $-2.4 \cdot 10^{-2}$ respectively. The prime block's contribution is negative; the positive energy is supplied by the archimedean block and partly consumed by the boundary. This is a single window and a single direction, and only the sign count n_- is invariant — the size of $\lambda_{\min}$ is not a margin, since structural nulls make it vanish by design once $n > m$. With that caution, the measurement points where the classical partial results also point: the difficulty of the positivity sits at the archimedean place, not with $\Lambda \geq 0$.

10.5 The exchange rate: linear detection against exponential cancellation

The archimedean difficulty admits a quantitative statement in the Weil form itself, which we record because it fixes what any proof must overcome. Take $g = \psi * \tilde{\psi}$ with $\psi = \mathbf{1}_{[-U/2, U/2]}$, so

$g(x) = (U - |x|)_+$ and $\hat{g}(r) = 4 \sin^2(Ur/2)/r^2 \geq 0$, and write the explicit formula as $W = \text{Arch} + \text{Pole} - \text{Prime} = \sum_{\rho} \hat{g}(\gamma_{\rho})$. Two measurements, with the archimedean tail taken analytically ($\hat{g}$ decays as r^{-2} against a $\log r$ digamma factor, so truncation at $|r| \leq 400$ leaves an error comparable to W itself and must be corrected):

First, W is essentially constant while its constituents diverge. At $U = 0.549, 2, 4$ one finds $W = +0.085, +0.107, +0.070$ against $\text{Pole} = 0.61, 8.69, 44.20$ and $\text{Prime} = 0, 3.49, 30.15$. The value is independently confirmed by $W = \sum_{\rho} \hat{g}(\gamma_{\rho}) \rightarrow 2 \sum_{\gamma > 0} \gamma^{-2} \cdot 2 \approx 0.092$. So the Weil form is a difference of quantities growing like $e^{U/2}$ whose difference is bounded, and the relative margin W/Pole falls from $1.4 \cdot 10^{-1}$ to $1.6 \cdot 10^{-3}$ across that range.

Second, the detection reach is only linear. A planted off-line pair at $\gamma = \beta \pm i\delta$ contributes $2 \operatorname{Re} \hat{g}(\beta + i\delta) \sim \pm 2 \cosh(U\delta)/|\gamma|^2$, oscillating in sign with amplitude $e^{U\delta}$ against the background ≈ 0.09 ; numerically W first turns negative at

$$U_{\text{detect}} \approx 1.6/\delta \quad (\delta = 0.5, 0.3, 0.2, 0.1 \Rightarrow U = 3.51, 5.31, 7.99, 14.66),$$

with the all-real control staying positive at every U tested.

Together: *detection depth is bought linearly in the support, while the cancellation to be resolved grows exponentially in it*. The narrow-support regime in which Weil positivity is known unconditionally is not narrow by weakness of technique; at $U = (\log 3)/2$ the reach is $\delta \gtrsim 2.9$, outside the critical strip, so it excludes nothing, and sharpening constants moves the threshold only logarithmically. A proof must break this exchange rate rather than improve within it — which is the same discipline the appendix records below, where every estimate that discarded a cancellation failed by orders and every exact evaluation succeeded.

11 The operator reading: a self-adjoint generator for the functional-equation pairing, and A as its spectral determinant

Everything in this section is unconditional, and every statement carrying a `teletype` name is machine-checked (`RequestProject/FoliatedPolarization.lean`, `RequestProject/FoliatedDeterminant.lean`, `RequestProject/HilbertPolyaResolventTrace.lean`; Lean 4/Mathlib, zero sorry, axiom footprint `{propext, Classical.choice, Quot.sound}` on every cited theorem). The frame is Deninger’s [8]: a flow with closed orbits of lengths $\log p^k$ whose generator carries the zeros as spectrum, RH delivered — as for curves over finite fields — by positivity of a polarization for which the generator is self-adjoint. We construct no foliated space and no leafwise cohomology. What is constructed is the rest of that frame’s inventory — spectral coordinate, polarization, self-adjointness, trace, determinant — together with the observation that ties it to this paper: the central object A of §1 *is* the determinant.

The spectral coordinate. For a zero $\rho = \beta_{\rho} + i\gamma_{\rho}$ put

$$t_{\rho} := -i(\rho - \tfrac{1}{2}) = \gamma_{\rho} + i(\tfrac{1}{2} - \beta_{\rho}), \quad \rho = \tfrac{1}{2} + it_{\rho},$$

so $t_{\rho} \in \mathbb{R} \iff \beta_{\rho} = \tfrac{1}{2}$ (`spectralCoord_real_iff`). The functional-equation reflection $\rho^* := 1 - \bar{\rho}$ is an involution fixing exactly the critical line (`feReflect_eq_self_iff`), and it *conjugates* the coordinate:

$$t_{1-\bar{\rho}} = \overline{t_{\rho}} \quad (\text{spectralCoord_feReflect}). \quad (3)$$

In this coordinate the zero multiset of A is $\{t_\rho\}$: the real support points x_i of (1) are the on-line zeros, and the conjugate pairs $z_j, \bar{z}_j$ are the off-line FE pairs $\{\rho, 1 - \bar{\rho}\}$, landed on $\{t_\rho, \bar{t}_\rho\}$.

Proposition 47 (the polarization). *On the reflection block $\mathbb{C}e_\rho \oplus \mathbb{C}e_{\rho^*}$ with multiplicity weight $m > 0$ define*

$$\langle v, w \rangle_m := m(v_1 \bar{w}_2 + v_2 \bar{w}_1), \quad \Theta|_{\text{block}} := \text{diag}(t_\rho, \bar{t}_\rho),$$

the diagonal action being forced by (3). Unconditionally:

- (i) $\langle \Theta v, w \rangle_m = \langle v, \Theta w \rangle_m$ for every zero, on the line or off it (**blockOp-selfAdjoint**);
- (ii) off the line ($\rho^* \neq \rho$) the block is hyperbolic of signature $(1, 1)$: the pairing is positive on $e_\rho + e_{\rho^*}$ and negative on $e_\rho - e_{\rho^*}$ (**pairBlock-indefinite**);
- (iii) on the line ($\rho^* = \rho$) the block degenerates to the single line $\mathbb{C}e_\rho$, on which the pairing is the positive norm $2m|z|^2$ (**pairForm-online-pos**).

Summing blocks over the multiset polarizes the zero space as a Krein space on which the generator is self-adjoint; the negative index is the number of off-line FE pairs — on a window W , exactly the census $q(W)$ of Theorem 2, which is thereby the signature computation of this pairing, window by window; the pairing itself is the FE contraction of the trichotomy of Remark 37. Hence

$$\text{RH} \iff \text{the pairing is definite} \iff \text{the Krein space is a Hilbert space},$$

which is Hypothesis 20 in operator coordinates: by the rigidity discussion of §9 this supplies a generator for the pairing, not a new reduction.

A calibration, itself compiled. A self-adjoint operator on a Pontryagin space with κ negative squares has at most κ pairs of non-real eigenvalues (the Krein–Langer bound). Here that bound is saturated identically: per block, non-real spectrum and indefiniteness are the same event, $\text{Im } t_\rho \neq 0 \iff \rho^* \neq \rho$ (**block_nonreal_iff_indefinite**), so the abstract theorem returns the census and nothing else. It becomes informative only when κ is bounded by an input external to the spectrum; in the function-field analogue that input is the Hodge index theorem on $C \times C$. Nothing in this paper supplies such a bound; the anchored transport of §9 is this paper’s candidate for the role.

The trace. Unconditionally (**hilbertPolya_resolvent_trace**; net-limit form **hilbertPolya_resolvent_trace**) for w, w_0 whose chart points $\frac{1}{2} + iw, \frac{1}{2} + iw_0$ avoid the zero set, the two-point resolvent sum over the actual zero multiset, weighted by the analytic multiplicities m_ρ , converges absolutely and equals the completed logarithmic derivative read in the chart:

$$\sum_{\rho} m_{\rho} \left[(t_{\rho} - w)^{-1} - (t_{\rho} - w_0)^{-1} \right] = -i \left[\frac{\xi'}{\xi} \left(\frac{1}{2} + iw \right) - \frac{\xi'}{\xi} \left(\frac{1}{2} + iw_0 \right) \right]. \quad (4)$$

Two-point differencing kills the Hadamard constant and the $1/\rho$ normalizers exactly; no regularization convention survives into the statement.

Theorem 48 (A is the spectral determinant). *Put $\det_{\Theta}(w) := \xi(\frac{1}{2} + iw)$, so that $\det_{\Theta}(w) = A(-w) = A(w)$ by the functional equation. Unconditionally:*

- (i) (zero set = spectrum) $\det_{\Theta}(w) = 0 \iff w = t_{\rho}$ for some nontrivial zero ρ (`spectralDet_eq_zero_iff_spectrum`);
- (ii) (multiplicities) at every $w_0 \in \mathbb{C}$ the analytic vanishing order of $\det_{\Theta}$ equals the order of ξ at $\frac{1}{2} + iw_0$; at $w_0 = t_{\rho}$ it is m_{ρ} (`spectralDet_analyticOrderNatAt`, `spectralDet_orderNat_spectralCoord`);
- (iii) (canonical product) there are constants $c_1, c_0 \in \mathbb{C}$ with

$$\det_{\Theta}(w) = e^{c_1 w + c_0} \prod_{\rho} E_1\left(\frac{\frac{1}{2} + iw}{\rho}\right)^{m_{\rho}}, \quad E_1(u) = (1 - u)e^u,$$

the multiplicity-indexed Weierstrass product over the zero multiset, multipliable locally uniformly on $\mathbb{C}$ (`spectralDet_hadamard_factorization`, `xiProductMult`); each factor vanishes exactly at $w = t_{\rho}$;

- (iv) (trace identity) the left side of (4) equals $-\left[\det'_{\Theta} / \det_{\Theta}(w) - \det'_{\Theta} / \det_{\Theta}(w_0)\right]$ (`resolvent_trace_eq_neg_L`); the genus-one ambiguity $e^{c_1 w + c_0}$ cancels in every two-point difference.

The name is earned in the only sense the frame requires: A is an entire function whose zero multiset, with multiplicity, is the spectrum of the generator; it factors as the canonical genus-one product over that multiset; and its two-point logarithmic derivative is the resolvent trace. Of the two services a leafwise cohomology would render, the finiteness of the trace is done by the product; the positivity is not, and is Hypothesis 20, i.e. the seat. Read backwards, the identification explains why the window moments carry the program: by (1) the $\mu_k(W)$ are the contour moments of $d \log A = d \log \det_{\Theta}$ — the power sums of the generator's spectrum localized to the window — and the census of Theorem 5 is the generator's spectral resolution read window by window.

11.1 The seat criterion split, and a certified channel

Everything in this subsection is unconditional and machine-checked (`RequestProject/DVPLadder.lean`, `DVPChannelBase.lean`, `DVPLadder126.lean` and their dependency closure; Lean 4/Mathlib, zero `sorry`, axiom footprint `{propext, Classical.choice, Quot.sound}` on every cited theorem). For an anchor $w \in \mathbb{C}$ with $\frac{1}{2} + iw$ and $\frac{1}{2} + i\bar{w}$ off the zero set and $w \neq \bar{w}$, write

$$F(w) := \operatorname{Re} \sum_{\rho} m_{\rho} (t_{\rho} - w)^{-1} (t_{\rho} - \bar{w})^{-1},$$

the FE-symmetric two-point quadratic form of the resolvent of (4) — per reflection block, exactly the pairing of Proposition 47 evaluated at the anchor pair $\{w, \bar{w}\}$.

The criterion. There is an explicit $c > 0$ such that, as a compiled equivalence (`seat_criterion_channel_iff`),

$$\text{RH} \iff F(w) \geq 0 \text{ for every admissible anchor in the channel } e^4 + 2 \leq |\operatorname{Re} w|, \quad |\operatorname{Im} w| < \frac{1}{2} - \frac{c}{\log(|\operatorname{Re} w| + \frac{1}{2})}$$

This is Hypothesis 20 localized: definiteness of the polarization need only be checked on a shallow channel hugging the real axis, at height growing like $\frac{1}{2} - O(1/\log)$. The criterion as originally compiled carries a second, finite conjunct — every zero of ordinate $|\gamma| < e^4 + \frac{5}{2}$ on the line — and that conjunct is now a theorem (`conjunct_one`), so the split collapses to the channel alone.

The certified ladder, and precisely what it settles. The finite conjunct asks only for the zeros of ordinate below $e^4 + \frac{5}{2} \approx 57.1$, and it is *already discharged* by the first rung: `online_below_58` covers that range, so conjunct (1) of the split is closed and no further height is required for it. Every rung above 58 therefore serves conjunct (2) alone, by widening the window on which the transfer `seat_channel_of_online_range` gives channel positivity. We record this explicitly because it delimits the ladder's reach: that window is bounded at every stage, while conjunct (2) quantifies over an unbounded set of heights, so no finite number of rungs closes it. What the ladder yields is a certified floor, not a route to the closure. With that said, the ladder is at present at height $743/2$ (`online_below_743o2`): every nontrivial zero with $|\gamma| \leq \frac{743}{2}$ has $\beta = \frac{1}{2}$. The census $N(743/2) = 183$ is itself a compiled count (`stripZeroCountMult_743o2`) — obtained by lifting the argument principle along the standard contour with kernel-checked rational enclosures, at what is itself a half-integer anchor height: every integer anchor in $[140, 230]$ fails the segment floors, and the anchor scan has selected half-integer heights ever since (an integer anchor T is carried in half form as $2T/2$); the 183 zeros are then separated by 184 certified sign changes of the real detector $\text{Re } \Lambda(\frac{1}{2} + it)$ (`hardyG`) at stations including seventeen half-integer heights and four quarter-integer heights (the first, $\frac{741}{4}$, placed at the Hardy-function peak of the thinnest zero gap, certified at $N = 981$; the deepest, $\frac{1181}{4}$, at $N = 1989$), forced by close pairs inside one integer cell or by thin margins at the neighbouring integers. Where a close pair is too narrow for any dyadic station to clear the first-order Euler–Maclaurin cost, the certificate switches instrument rather than deepening the sum: the second-order enclosure (`zeta_EM_enclosure_second_numeric`), which books the B_2 term into the head instead of the error, reduces that cost by a factor $\approx 63/N$ and settles such a station at a table depth already available. Each station is an exact-rational Euler–Maclaurin certificate; where the crude boundary bound $\sqrt{N}/t$ would consume the margin, the boundary term is instead evaluated in closed form and contributes with its true sign (`phase_station_lower_half_eval`). The construction is a mechanized rung procedure and has been run nineteen times ($58 \rightarrow 63 \rightarrow 74 \rightarrow 81 \rightarrow 91 \rightarrow 109 \rightarrow 126 \rightarrow \frac{289}{2} \rightarrow \frac{309}{2} \rightarrow \frac{399}{2} \rightarrow \frac{435}{2} \rightarrow \frac{471}{2} \rightarrow \frac{525}{2} \rightarrow \frac{561}{2} \rightarrow \frac{581}{2} \rightarrow \frac{618}{2} \rightarrow \frac{671}{2} \rightarrow \frac{687}{2} \rightarrow \frac{743}{2}$); nothing in it is height-specific beyond the witness tables.

The transfer. On-line-ness up to height T transfers to channel positivity on the matching window (`seat_channel_of_online_range`): if every zero with $|\gamma| \leq T$ is on the line, then $F(w) \geq 0$ for every admissible anchor with $|\text{Re } w| \leq T - \frac{1}{2}$ — at *every* height $|\text{Im } w|$, not merely in the channel. With the ladder this is unconditional to $|\text{Re } w| \leq 371$ (`seat_channel_to_371`).

Exact status. The compiled equivalence and the compiled transfer bracket the remaining content precisely: positivity of F on the channel window $|\text{Re } w| \leq T - \frac{1}{2}$ is a theorem for $T = 743/2$ and is extensible rung by rung, while positivity on the full channel is, by the criterion, the conclusion itself — the two statements differ exactly by the passage $T \rightarrow \infty$, and the equivalence $\text{RH} \iff \text{on-line above } 58$ (`rh_iff_online_above_58`) shows no finite extension closes it. What the split changes is the shape of the obligation: the seat of §9 now has a certified floor and a single quantitative frontier, and every rung moves the frontier by a compiled theorem.

The residual conjunct, as a sign condition. The remaining conjunct admits a form with no sum over zeros in it. Writing $w = x + iy$, the two chart points of the seat energy are $s_1 = \frac{1}{2} - y + ix$ and $s_2 = \frac{1}{2} + y + ix = 1 - \bar{s}_1$, so the functional equation $\xi(1 - s) = \xi(s)$ and the reality $\xi(\bar{s}) = \overline{\xi(s)}$, differentiated, give $\xi'/\xi(s_2) = -\xi'/\xi(\bar{s}_1)$. The two-point difference is then

twice a real part, and dividing by $w - \bar{w} = 2i \operatorname{Im} w$ leaves (`seat_energy_re_eq`, unconditional)

$$F(w) = - \frac{\operatorname{Re}[\xi'/\xi(\frac{1}{2} + iw)]}{\operatorname{Im} w}.$$

Since conjunct (1) is discharged at height 58, the whole of RH is therefore the sign condition

$$\operatorname{Re}[\xi'/\xi(\sigma + it)] \leq 0 \quad (\sigma < \tfrac{1}{2})$$

at admissible points. The same antisymmetry makes $\operatorname{Re}[\xi'/\xi]$ vanish identically on $\sigma = \frac{1}{2}$; since ξ is entire, this function is harmonic off the zeros, which are its only singularities, and it tends to $-\infty$ as $\sigma \rightarrow -\infty$. For $\sigma < 0$ the condition is unconditional — there $1 - s$ lies in the half-plane of absolute convergence, so $\zeta'/\zeta(1 - s)$ is bounded and $\operatorname{Re}[\xi'/\xi] \approx -\frac{1}{2} \log(|t|/2\pi)$ — which is the exterior positivity already recorded. The content is the strip $0 < \sigma < \frac{1}{2}$, where the same reflection sends $1 - s$ into $(\frac{1}{2}, 1)$ and ζ'/ζ is no longer bounded. We state this because it is the sharpest compiled form of the obligation, not because it weakens it: the reformulation is an equivalence, and a single zero off the line defeats the inequality, the offending pair contributing $2y[(a^2 - y^2) - d^2]/D$ with $a = |\beta - \frac{1}{2}|$, $d = t - \gamma$, which is positive exactly on the disc $d^2 + y^2 < a^2$ and unbounded as $y \rightarrow a^-$.

The bank read in order. The sign condition is posed inside the strip, where the Dirichlet series does not converge, so it is worth recording in what sense the bank is still available there. The carrier's alternating bank $\sum_{n \geq 0} (-1)^n (n+1)^{-s}$, summed *in the order its phasors enter*, converges on all of $\operatorname{Re} s > 0$: consecutive phasors differ by

$$\|a^{-s} - (a+1)^{-s}\| \leq \|s\| a^{-\sigma-1} \quad (a > 0, \sigma \geq -1),$$

by the fundamental theorem of calculus on $[a, a+1]$ (`cpow_diff_norm_le`), so the paired series $\sum_k [(2k+1)^{-s} - (2k+2)^{-s}]$ is absolutely summable exactly when $\sigma > 0$ (`paired_bank_summable`), and the ordered partial sums converge there (`etaPartial_tendsto`). The limit is holomorphic on that half-plane, and on it

$$\lim_{N \rightarrow \infty} \sum_{n < N} \frac{(-1)^n}{(n+1)^s} = (1 - 2^{1-s})\zeta(s) \quad (\operatorname{Re} s > 0, s \neq 1),$$

proved by the identity theorem from $\operatorname{Re} s > 1$ across the punctured half-plane $\{\operatorname{Re} s > 0\} \setminus \{1\}$, which is covered by four convex opens and is therefore preconnected (`etaLim_differentiableOn`, `etaLim_eq_zeta`; both unconditional, standard axioms).

We are explicit about what this does and does not add. It does *not* enlarge the region in which ζ is available unconditionally in this development: the Euler–Maclaurin representation `truncated_zeta_pos_re` already covers $\operatorname{Re} s > 0, s \neq 1$. What changes is the summation order and the shape of the error: the truncation error after N ordered terms is the tail of a single elementary majorant, rather than a tail of fractional-part integrals. Nor does it bear on the sign condition, which is an obligation over an unbounded range of t and is not discharged by any pointwise representation of ζ .

12 Appendix: the formalized theta-kernel Turán program

Parallel to the transport program of the body, a machine-checked campaign (`RequestProject/ThetaWronskianSu` Lean 4/Mathlib, about 9,800 lines, zero `sorry`, axiom footprint `{propxt, Classical.choice, Quot.sound}`)

on every theorem cited below) attacks the same moment inequalities through Riemann’s theta kernel

$$K(u) = 4 \sum_{n \geq 1} (2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}) e^{-\pi n^2 e^{2u}},$$

whose half-line moments $m_k = \int_0^\infty u^k K(u) du$ are the paper’s central-coefficient data.

Attribution, and the status of this branch. The analytic content of this program is classical; the formalization is what is new. Monotonicity of the radial hazard $-K'(u)/(2uK(u))$ — equivalently concavity of $\log K(\sqrt{t})$, which is the gate below — is Csordas–Varga [10], Theorem 2.1. The Turán inequalities for the Maclaurin coefficients of Ξ , solving a problem of Pólya from 1927, are Csordas–Norfolk–Varga [9]; the central case $3m_2^2 > m_0m_4$ used below is their $m = 1$ — the one case their analytic argument did not reach, established in 1986 by certified high-precision computation and analytically in [10] — with margin 6.96% ($m_0m_4/m_2^2 = 2.7911$ against 3). Hyperbolicity of the Jensen polynomials $J^{d,n}$ for *every* n is known for all $d \leq 8$ by Griffin–Ono–Rolen–Zagier [11], which subsumes the five Theta–Wronskian levels obtained here. Since Jensen’s criterion already lives at $n = 0$ (Ξ lies in the Laguerre–Pólya class, i.e. RH holds, iff $J^{d,0}$ is hyperbolic for every d), the tower at the centre is not a subfamily of the criterion but the criterion itself, its rungs climbed by successively higher-order positivity of the kernel. This section therefore contributes a machine-checked development of that classical circle, not new arithmetic. It is worth stating what this does *not* establish: no barrier is known to the degree ladder. Jensen’s criterion already lives at $n = 0$ — Ξ lies in the Laguerre–Pólya class, equivalently RH holds, iff $J^{d,0}$ is hyperbolic for every d — so finite-degree results are progress along the criterion itself rather than a ceiling. (One tempting argument in the other direction is spurious: Pólya-frequency membership of Φ fails, but by Schoenberg that would require the transform of Φ to be the reciprocal of a Laguerre–Pólya function, whereas it is Ξ itself; the failure records only that Ξ has zeros, and carries no information about $\Xi \in LP$.) The estimates developed here are moreover independent of d : they control the sum over the theta index, while the degree selects which moment inequality is the target.

The compiled results, at exactly their proved strength:

1. **Analytic layer.** Term-by-term first and second differentiation of the theta series on $u > 0$ under summable exponential majorants (`riemannXiKernel_hasDerivAt`, `riemannXiKernel_deriv_hasDerivAt`); positivity, continuity, and integrability of all polynomial moments.
2. **The radial hazard and its gate.** For $H(u) = -K'(u)/(2uK(u))$ the exact identity $H'(u) = N(u)/(2u^2K(u)^2)$ is formalized, with $N(u) = u(K')^2 - uKK'' + KK'$ (`xiRadialHazard_hasDerivAt`); positivity of N on $u > 0$ is equivalent, in compiled form, to domination of a weighted slope variance by a mass-times-gate term (`xiRadialConcavityNumerator_posDef`).
3. **The consequence chain.** Integration by parts gives the exact hazard-moment identities $\int u^2KH = m_0/2$ and $\int u^4KH = 3m_2/2$ (`xiRadialHazard_weight_two/four`); the covariance double integral evaluates unconditionally to $3m_2^2 - m_0m_4$ (`xiRadialCovarianceIntegral_eq`); and positivity of N therefore forces the strict central Turán inequality $m_0m_4 < 3m_2^2$ and closes Theta–Wronskian levels four and five, hence levels one through five (`theta_wronskian_levels_one_to_five`). Levels one through three are unconditional (`riemann.theta_wronskian_levels_one_to_three`), as is positive definiteness of every finite Hankel moment matrix (`xiMomentMatrix_posDef`).

4. **The hazard-moment identities are generating, and yield a uniform family.** The identities of item 3 are not isolated: they are the $k = 2, 4$ cases of

$$2 \int_0^\infty u^k K H = \frac{k-1}{2} m_{k-2} \quad (k \geq 2),$$

verified to ten digits for $k \leq 10$, and the covariance evaluation is their antisymmetrisation rather than an independent fact: with weight $w = u^2 K$,

$$\iint (u^2 - v^2)(H(u) - H(v))w(u)w(v) = 2 \left\{ \frac{3m_2}{4} \frac{m_2}{2} - \frac{m_4}{2} \frac{m_0}{4} \right\} = \frac{1}{4} (3m_2^2 - m_0 m_4).$$

Shifting the weight to $w = u^{2j} K$ keeps the antisymmetrisation at H^0 and H^1 , so the same two identities supply every ingredient and no H^2 -moment ever appears. This gives, for each $j \geq 1$,

$$C_j = \frac{1}{4} \left[(2j+1) m_{2j}^2 - (2j-1) m_{2j-2} m_{2j+2} \right],$$

with C_1 the central Turán expression. The integrand $(u^2 - v^2)(H(u) - H(v))w(u)w(v)$ is pointwise nonnegative for *every* j as soon as H is increasing, so a single proved gate delivers the whole family $C_j > 0$ at once, not one rung. Measured positive for $j \leq 6$, with ratios 1.0748, 1.0627, 1.0543, 1.0481, 1.0432, 1.0394 decreasing towards 1.

We record two dead ends here so they are not re-walked. Enlarging the Andreief determinant $\det[\int u^{2(i-1)} H^{j-1} w]$ instead of shifting the weight is the wrong generalisation: it forces H^j -moments with $j \geq 2$, which are Fisher-information functionals and do not reduce by parts (a natural candidate misses by 6–9%); and it mismatches degrees, being degree d in the moments where the Jensen discriminant at degree d is degree $2(d-1)$, the two agreeing only at $d = 2$. Second, the family C_j consists of Laguerre–Turán inequalities at every index; these are *necessary* for Ξ to lie in the Laguerre–Pólya class but not sufficient, so the uniformity here is uniformity across indices, not a route to hyperbolicity of $J^{d,0}$ for all d .

5. **State of the gate.** The ground-index gate $s_0(u) - u c_0(u) > 0$ is proved on the whole half-line, the thin region closed by a two-sided exponential window (quadratic Taylor minorant, squared Padé majorant) against four-digit bounds for π (`xiTermGate_zero_pos`). At the origin the kernel’s evenness forces N to vanish to third order, $N(u) = au^3 + O(u^5)$ with $a = K''(0)^2 - K(0)K''''(0)/3$; the strict inequality $3K''(0)^2 > K(0)K''''(0)$ is compiled from certified partial-sum bounds ($S_0 < 9/20$, $S_2 < -8333/1000$, $S_4 < 410$; `xiOriginThetaConstant_pos`, `xiOriginCubicCoefficient_pos`), with measured margin $a \approx 152.2$, about 14% relative.

The gate is now proved unconditionally on the whole of $[1/4, \infty)$ (`xiRadialConcavityNumerator_pos_of` for π) extending the earlier $[3/10, \infty)$ result by one notch in the per-term decay rate, $18 \rightarrow 16$: at $u \geq 1/4$ the ground coordinate satisfies $q_0 = \pi e^{2u} \geq 5.1$, the gap estimate $20n \leq 4nq_0$ still holds, and $(n+2)^4 \leq 16e^{4n}$ converts $2(n+2)^4 e^{-20n}$ into $32e^{-16n}$. The derived rates fall in step (q -weighted 14, squared-slope 12) and every prefactor survives, each being a geometric sum $\sum e^{-cn} \leq 2$ valid for all $c \geq 12$. The one place the notch bites is $4q_0^2 \leq e^{q_0}$, which needs seven Taylor terms at $q_0 \geq 5.1$ where six sufficed at $q_0 \geq 5.7$.

6. **What remains on $(0, 1/4]$.** Termwise estimation of N fails there: near the origin N is a cancellation of order 10^3 between its three summands, and bounding them separately

loses a factor ≈ 22 (measured: a termwise majorant gives $|N^{(5)}| \leq 7.83 \times 10^6$ against a true 3.60×10^5). The remedy is to work with the hazard rather than with N , which the compiled identity of item 2 already supplies: writing $g = (\log K)' - u(\log K)'' = 2u^2 H'$, the identity $H' = N/(2u^2 K^2)$ reads $N = K^2 g$, so $K^2 > 0$ factors out and the gate is *equivalent* to $g > 0$, i.e. to monotonicity of the radial hazard. This is a restatement of `xiRadialHazard_hasDerivAt`, not a new identity; it is recorded here only because the g form makes the next step visible. That step is elementary: $g(0) = 0$ and

$$g'(u) = -u(\log K)''',$$

so the sign condition $(\log K)''' < 0$ on an interval makes g rise from zero and the gate hold there — by integration, with no Taylor remainder and no jet coefficients. Near the origin that sign is governed by the constant already compiled above: the quartic log-coefficient is $L_4 = a_4/a_0 - a_2^2/(2a_0^2) = -a/(8K(0)^2)$ with a the cubic coefficient, so $(\log K)''' < 0$ near 0 is exactly $a > 0$. Numerically $(\log K)'''$ is negative throughout the range examined ($u \leq 3$), with small- u slope $24L_4 = -143.0$; this is measured, not proved. Remaining for the gate: the sign of $(\log K)'''$ as a proved statement, and the formalized evenness of K (Jacobi inversion; the one-dimensional Poisson ingredient is compiled elsewhere in the repository). Three majorant strategies were refuted by measurement and are recorded as such in the source — Cauchy coefficient estimates (loose by 2×10^4 , because the Taylor coefficients of K are small precisely by cancellation), an order-nine Lagrange remainder (loose by 10), and an ℓ^1 bound on the derivative polynomials (loose by 2×10^5).

The last of these is a defect of the instantiation, not of the route, and the repair is worth recording because it is the same repair three times over. Writing $D^k T_n = 4e^{u/2} P_k(q) e^{-q}$ with $P_k = \sum_j \binom{k}{j} 2^{j-k} R_j$ a *single* polynomial — the R_j obeying $R_{j+1} = 2q(R'_j - R_j)$ — and evaluating P_k rather than summing $|R_j|$ recovers almost everything: the majorants for $|K^{(4)}|, \dots, |K^{(7)}|$ fall from 1.095, 2.434, 1.271, 4.242 times the true value to 1.001, 1.001, 1.051, 1.503. With that, the two-term criterion closes on $(0, 1/20]$ with margin 1.60, where the ℓ^1 form had been short by a factor 3 to 5.

Two further compiled ingredients on this route. The cumulant identity

$$(\log K)''' = \langle c' \rangle + 3 \text{Cov}(s, c) + \mu_3(s), \quad p_n = T_n/K,$$

verified symbolically, exhibits the mechanism: near the origin the sign is carried by $\mu_3(s) < 0$, the log-slopes being violently left-skewed ($s_0 = 0.044$ against $s_1 = -20.36$), while past $u \approx 0.048$ the first summand turns negative and dominates as $p_0 \rightarrow 1$. Its termwise input is proved: `xiKernelTermLogCurvatureDeriv_neg` gives $c'_n < 0$ at every index for $u \geq 1/10$, since $c'_n = 8q(6(2q+3)/I^3 - 1)$ is negative exactly when $8q^3 - 36q^2 + 42q - 45 > 0$, which holds on $[19/5, \infty)$ and hence at every index because $q_n \geq q_0 \geq 19/5$.

7. **The Suzuki–Gram side.** The finite Suzuki Gram inequality, signed Fredholm solvability and identification, and the corrected-pencil Galerkin transfer are compiled with their two analytic inputs isolated as hypotheses — an operator contraction and the identification of the analysis integrals with a contraction of the Weil vectors (`xiSuzukiGram_inequality`, `xiGalerkin_fredholm_signed_identification`, `xiCorrectedPencil_galerkin_transfer`). The contraction is discharged unconditionally at a certified corrected-pencil level: the single-node Galerkin compression of the actual kernel at node three satisfies $\|A\| = K(9) < 1$ by a compiled decay bound and elementary bounds for e (`xiSuzukiGalerkin_contraction_example`), so the signed Fredholm chain fires there with no hypothesis (`xiGalerkin_fredholm_example`).

The two campaigns are complementary reductions of the same object: the body's window census reads the zeros through contour Hankels and isolates the seat; the formalized program reads the central coefficients through the theta kernel and isolates the numerator gate and the Suzuki contraction. Each residue is stated with its consequence chain fully compiled above it.

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